# 1 Equations in $\rho$ coordinates

$$0 = \frac{1}{V'}\frac{\partial}{\partial\rho}\left[V'\langle|\nabla\rho|^2\rangle\frac{\partial}{\partial\rho}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_\mathrm{e} + Z_\mathrm{i}en_\mathrm{i} + Z_\mathrm{z}en_\mathrm{z} + Z_\mathrm{b}en_\mathrm{b} + Z_\mathrm{b}egn_\mathrm{b}^\mathrm{rp}\right) \tag{1.1}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi}_t = {} & V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\left[\frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\psi_t\right] \\ & + \frac{\mu_0}{\langle B^\theta\rangle}\left[-en_\mathrm{e}\left(\langle Bu_{\mathrm{e}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle\right) + Z_\mathrm{i}en_\mathrm{i}\left(\langle Bu_{\mathrm{i}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle\right)\right. \\ & \left. + Z_\mathrm{z}en_\mathrm{z}\left(\langle Bu_{\mathrm{z}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right) + Z_\mathrm{b}en_\mathrm{b}\left(\langle Bu_{\mathrm{b}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{1.2}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi} = {} & \frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\left[V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\psi\right] \\ & - \frac{\mu_0}{\langle R^{-2}\rangle}\left(-en_\mathrm{e}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle + Z_\mathrm{i}en_\mathrm{i}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle + Z_\mathrm{z}en_\mathrm{z}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + Z_\mathrm{b}en_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\end{aligned} \tag{1.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{1.4}$$

$$\frac{\partial}{\partial t}\psi_t = \dot{\psi}_t \tag{1.5}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V'n_\mathrm{e}) = {} & -\frac{1}{V'}\frac{\partial}{\partial\rho}[V'n_\mathrm{e}(u_\mathrm{e}^\rho - u_g^\rho)] + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\ & + Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e} - n_\mathrm{esup}) + (1 - f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{1.6}$$

$$\begin{aligned}0 = {} & -(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}n_\mathrm{e}T_\mathrm{e} + en_\mathrm{e}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}\Phi \\ & - e\frac{\partial\psi}{\partial\rho}In_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + e\frac{\partial\psi}{\partial\rho}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle\end{aligned} \tag{1.7}$$

$$\begin{aligned}m_\mathrm{e}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\parallel}\rangle = {} & -\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}} + \hat{\mu}_2^\mathrm{e}BV_{2\mathrm{e}} \\ & + \ell_{11}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\parallel}\rangle \\ & - \ell_{12}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\ & - en_\mathrm{e}\langle BE_\parallel\rangle - \nu_\mathrm{L}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \nu_{0\mathrm{e}}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle \\ & - \frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\\
&-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle-\frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\\
&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right] \qquad (1.8)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V'm_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle)=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle+(u_{\mathrm{e}}^{\rho}-u_g^{\rho})m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\langle Ru_{\mathrm{e}\zeta}\rangle+\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\right]\\
&+\ell_{11}^{\mathrm{ee}}\langle Ru_{\mathrm{e}\zeta}\rangle+\ell_{11}^{\mathrm{ei}}\langle Ru_{\mathrm{i}\zeta}\rangle+\ell_{11}^{\mathrm{ez}}\langle Ru_{\mathrm{z}\zeta}\rangle+\ell_{11}^{\mathrm{eb}}\langle Ru_{\mathrm{b}\zeta}\rangle\\
&-\ell_{12}^{\mathrm{ee}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle-\ell_{12}^{\mathrm{ei}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle-\ell_{12}^{\mathrm{ez}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle\\
&-en_{\mathrm{e}}\langle RE_{\zeta}\rangle-e\frac{\partial\psi}{\partial\rho}n_{\mathrm{e}}u_{\mathrm{e}}^{\rho}\\
&-\frac{e\langle|\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}}\frac{\partial\psi}{\partial\rho}\left[(1-C_{\mathrm{T}})T_{\mathrm{e}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}+C_{\mathrm{T}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}+n_{\mathrm{e}}T_{\mathrm{e}}\frac{\partial\psi}{\partial\rho}C_{\mathrm{p}}\right]\\
&-\nu_{\mathrm{L}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle-\nu_{0\mathrm{e}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle \qquad (1.9)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{1}{V'^{5/3}}\frac{\partial}{\partial t}(V'^{5/3}n_{\mathrm{e}}T_{\mathrm{e}})=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[-\chi_{\mathrm{e}}\langle|\nabla\rho|^2\rangle n_{\mathrm{e}}\frac{\partial}{\partial\rho}T_{\mathrm{e}}+V_{\mathrm{e}}^{\mathrm{hp}}\langle|\nabla\rho|\rangle n_{\mathrm{e}}T_{\mathrm{e}}+\frac{5}{2}n_{\mathrm{e}}T_{\mathrm{e}}(u_{\mathrm{e}}^{\rho}-u_g^{\rho})\right]\\
&-u_g^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}-u_{\mathrm{i}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{i}}T_{\mathrm{i}}-u_{\mathrm{z}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{z}}T_{\mathrm{z}}-\hat{u}_{\mathrm{i}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{i}}\rangle-\hat{u}_{\mathrm{z}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle\\
&-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{i}}}{\tau_{\mathrm{ei}}}-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{z}}}{\tau_{\mathrm{ez}}}\\
&-\dot{\psi}_t\langle B^{\theta}\rangle(-en_{\mathrm{e}}\hat{u}_{\mathrm{e}\theta}+Z_{\mathrm{i}}en_{\mathrm{i}}\hat{u}_{\mathrm{i}\theta}+Z_{\mathrm{z}}en_{\mathrm{z}}\hat{u}_{\mathrm{z}\theta})\\
&+\dot{\psi}\left(-en_{\mathrm{e}}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle+Z_{\mathrm{i}}en_{\mathrm{i}}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle+Z_{\mathrm{z}}en_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right)\\
&-\nu_{\mathrm{L}}T_{\mathrm{e}}(n_{\mathrm{e}}-n_{\mathrm{esup}})-\frac{3}{2}\nu_{\mathrm{L}T_{\mathrm{e}}}n_{\mathrm{e}}(T_{\mathrm{e}}-T_{\mathrm{esup}})\\
&-E_{\mathrm{H}}\nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\
&+f_{\mathrm{ei}}P_{\mathrm{b}}+P_{\alpha\mathrm{e}}+P_{\mathrm{RFe}}-P_{\mathrm{Br}} \qquad (1.10)
\end{aligned}
$$

$$
\frac{5}{2}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle=-\hat{\mu}_2^{\mathrm{e}}\langle Bu_{\mathrm{e}\parallel}\rangle-\hat{\mu}_3^{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle+\hat{\mu}_2^{\mathrm{e}}BV_{1\mathrm{e}}+\hat{\mu}_3^{\mathrm{e}}BV_{2\mathrm{e}}
$$

$$
\begin{aligned}
&- \ell_{21}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\|}\rangle - \ell_{21}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\|}\rangle - \ell_{21}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\|}\rangle - \ell_{21}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\|}\rangle \\
&+ \ell_{22}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \ell_{22}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \ell_{22}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle
\end{aligned} \tag{1.11}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\mathrm{i}}) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V' n_{\mathrm{i}}(u_{\mathrm{i}}^{\rho} - u_g^{\rho})] + \nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\
&- \nu_{\mathrm{L}}(n_{\mathrm{i}} - n_{\mathrm{isup}}) + \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}}\frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}
\end{aligned} \tag{1.12}
$$

$$
\begin{aligned}
0 = &-(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} - Z_{\mathrm{i}} e n_{\mathrm{i}}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho}\Phi \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho} I\langle Bu_{\mathrm{i}\|}\rangle - Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho}\langle B^2\rangle\langle Ru_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.13}
$$

$$
\begin{aligned}
m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial t}\langle Bu_{\mathrm{i}\|}\rangle = &-\hat{\mu}_1^{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \hat{\mu}_2^{\mathrm{i}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\mu}_1^{\mathrm{i}} BV_{1\mathrm{i}} + \hat{\mu}_2^{\mathrm{i}} BV_{2\mathrm{i}} \\
&+ \ell_{11}^{\mathrm{ie}}\langle Bu_{\mathrm{e}\|}\rangle + \ell_{11}^{\mathrm{ii}}\langle Bu_{\mathrm{i}\|}\rangle + \ell_{11}^{\mathrm{iz}}\langle Bu_{\mathrm{z}\|}\rangle + \ell_{11}^{\mathrm{ib}}\langle Bu_{\mathrm{b}\|}\rangle \\
&- \ell_{12}^{\mathrm{ie}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \ell_{12}^{\mathrm{ii}}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \ell_{12}^{\mathrm{iz}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle BE_{\|}\rangle - \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}}\langle Bu_{\mathrm{b}\|}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&+ \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle - \frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right]
\end{aligned} \tag{1.14}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle) = &-\frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle + (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\langle Ru_{\mathrm{i}\zeta}\rangle + \langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle\right] \\
&+ \ell_{11}^{\mathrm{ie}}\langle Ru_{\mathrm{e}\zeta}\rangle + \ell_{11}^{\mathrm{ii}}\langle Ru_{\mathrm{i}\zeta}\rangle + \ell_{11}^{\mathrm{iz}}\langle Ru_{\mathrm{z}\zeta}\rangle + \ell_{11}^{\mathrm{ib}}\langle Ru_{\mathrm{b}\zeta}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&- \ell_{12}^{\mathrm{ie}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle - \ell_{12}^{\mathrm{ii}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle - \ell_{12}^{\mathrm{iz}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle RE_{\zeta}\rangle + Z_{\mathrm{i}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{i}} u_{\mathrm{i}}^{\rho} \\
&+ Z_{\mathrm{i}} \frac{e\langle |\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left[ (1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle - \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle \\
&- \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.15}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{i}} T_{\mathrm{i}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[ -\chi_{\mathrm{i}} \langle |\nabla\rho|^2\rangle n_{\mathrm{i}} \frac{\partial}{\partial \rho} T_{\mathrm{i}} + V_{\mathrm{i}}^{\mathrm{hp}} \langle |\nabla\rho|\rangle n_{\mathrm{i}} T_{\mathrm{i}} + \frac{5}{2} n_{\mathrm{i}} T_{\mathrm{i}} (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{e}}}{\tau_{\mathrm{ie}}} - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{z}}}{\tau_{\mathrm{iz}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{i}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{i}}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \langle B u_{\mathrm{b}\parallel}\rangle \frac{\langle B u_{\mathrm{i}\parallel}\rangle}{\langle B^2\rangle} + P_{\alpha\mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{1.16}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{i}} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle = &- \hat{\mu}_2^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel}\rangle - \hat{\mu}_3^{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \hat{\mu}_2^{\mathrm{i}} B V_{1\mathrm{i}} + \hat{\mu}_3^{\mathrm{i}} B V_{2\mathrm{i}} \\
&- \ell_{21}^{\mathrm{ie}} \langle B u_{\mathrm{e}\parallel}\rangle - \ell_{21}^{\mathrm{ii}} \langle B u_{\mathrm{i}\parallel}\rangle - \ell_{21}^{\mathrm{iz}} \langle B u_{\mathrm{z}\parallel}\rangle - \ell_{21}^{\mathrm{ib}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&+ \ell_{22}^{\mathrm{ie}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \ell_{22}^{\mathrm{ii}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \ell_{22}^{\mathrm{iz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle
\end{aligned} \tag{1.17}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} [V' n_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho})] + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 \\
&- \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}}
\end{aligned} \tag{1.18}
$$

$$
\begin{aligned}
0 = &- (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} - Z_{\mathrm{z}} e n_{\mathrm{z}} (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} \Phi \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{z}\parallel}\rangle - Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} \langle B^2\rangle \langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{1.19}
$$

$$
\begin{aligned}
m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel}\rangle = &- \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel}\rangle - \hat{\mu}_2^{\mathrm{z}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle + \hat{\mu}_1^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_2^{\mathrm{z}} B V_{2\mathrm{z}} \\
&+ \ell_{11}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{zi}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{zz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle B E_{\parallel} \rangle - \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left( \frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&+ \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left( \frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{\langle B^2 \rangle}{I} \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[ \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left( \langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \left( \langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right] \qquad (1.20)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[ \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \langle R u_{\mathrm{z}\zeta} \rangle + \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \right] \\
&+ \ell_{11}^{\mathrm{ze}} \langle R u_{\mathrm{e}\zeta} \rangle + \ell_{11}^{\mathrm{zi}} \langle R u_{\mathrm{i}\zeta} \rangle + \ell_{11}^{\mathrm{zz}} \langle R u_{\mathrm{z}\zeta} \rangle + \ell_{11}^{\mathrm{zb}} \langle R u_{\mathrm{b}\zeta} \rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle R \hat{q}_{\mathrm{e}\zeta} \rangle - \ell_{12}^{\mathrm{zi}} \langle R \hat{q}_{\mathrm{i}\zeta} \rangle - \ell_{12}^{\mathrm{zz}} \langle R \hat{q}_{\mathrm{z}\zeta} \rangle \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle R E_{\zeta} \rangle + Z_{\mathrm{z}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{z}} u_{\mathrm{z}}^{\rho} \\
&+ Z_{\mathrm{z}} \frac{e \langle |\nabla \rho|^2 \rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} \left[ (1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \qquad (1.21)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{z}} T_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[ - \chi_{\mathrm{z}} \langle |\nabla \rho|^2 \rangle n_{\mathrm{z}} \frac{\partial}{\partial \rho} T_{\mathrm{z}} + V_{\mathrm{z}}^{\mathrm{hp}} \langle |\nabla \rho| \rangle n_{\mathrm{z}} T_{\mathrm{z}} + \frac{5}{2} n_{\mathrm{z}} T_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{e}}}{\tau_{\mathrm{ze}}} - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{i}}}{\tau_{\mathrm{zi}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{z}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{z}}} n_{\mathrm{z}} (T_{\mathrm{z}} - T_{\mathrm{zsup}}) \\
&+ P_{\alpha \mathrm{z}} + P_{\mathrm{RFz}} \qquad (1.22)
\end{aligned}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle = &- \hat{\mu}_2^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\mu}_3^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_3^{\mathrm{z}} B V_{2\mathrm{z}} \\
&- \ell_{21}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel} \rangle - \ell_{21}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel} \rangle - \ell_{21}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel} \rangle - \ell_{21}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \ell_{22}^{\mathrm{ze}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \ell_{22}^{\mathrm{zi}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \ell_{22}^{\mathrm{zz}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle
\end{aligned}
$$

$$- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \tag{1.23}$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{\mathrm{b}}) = -\frac{1}{V'} \frac{\partial}{\partial \rho} [V' n_{\mathrm{b}} (u_{\mathrm{b}}^{\rho} - u_g^{\rho})] + \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} - g \frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} - \nu_{\mathrm{b}} n_{\mathrm{b}} - \nu_{\mathrm{bL}} n_{\mathrm{b}} - \nu_{\mathrm{eff}} g (\sqrt{\delta} n_{\mathrm{b}} - n_{\mathrm{b}}^{\mathrm{rp}})$$
$$+ \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle f_t D_{\mathrm{b}} \frac{\partial}{\partial \rho} n_{\mathrm{b}} \tag{1.24}$$

$$0 = -(\langle B^2 \rangle \langle R^2 \rangle - I^2) \frac{\partial}{\partial \rho} n_{\mathrm{b}} T_{\mathrm{b}} - Z_{\mathrm{b}} e n_{\mathrm{b}} (\langle B^2 \rangle \langle R^2 \rangle - I^2) \frac{\partial}{\partial \rho} \Phi$$
$$+ Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{b}\parallel} \rangle - Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} \langle B^2 \rangle \langle R u_{\mathrm{b}\zeta} \rangle \tag{1.25}$$

$$m_{\mathrm{b}} \frac{\partial}{\partial t} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle = -(\hat{\mu}_1^{\mathrm{b}} - \ell_{11}^{\mathrm{bb}}) \langle B u_{\mathrm{b}\parallel} \rangle + \sum_{a \neq \mathrm{b}} \ell_{11}^{\mathrm{b}a} \langle B u_{a\parallel} \rangle + \hat{\mu}_1^{\mathrm{b}} B V_{1\mathrm{b}} + Z_{\mathrm{b}} e n_{\mathrm{b}} \langle B E_{\parallel}^{A} \rangle$$
$$- \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle - \nu_{0\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle + \langle B S_{\mathrm{m}\parallel \mathrm{b}} \rangle \tag{1.26}$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle) = -\frac{1}{V'} \frac{\partial}{\partial \rho} [V' (u_{\mathrm{b}}^{\rho} - u_g^{\rho}) m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle]$$
$$+ \ell_{11}^{\mathrm{be}} \langle R u_{\mathrm{e}\zeta} \rangle + \ell_{11}^{\mathrm{bi}} \langle R u_{\mathrm{i}\zeta} \rangle + \ell_{11}^{\mathrm{bz}} \langle R u_{\mathrm{z}\zeta} \rangle + \ell_{11}^{\mathrm{bb}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle$$
$$+ Z_{\mathrm{b}} e n_{\mathrm{b}} \dot{\psi} + Z_{\mathrm{b}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{b}} u_{\mathrm{b}}^{\rho} - \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle$$
$$- \nu_{0\mathrm{i}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle + \frac{I}{\langle B^2 \rangle} \langle B S_{\mathrm{m}\parallel \mathrm{b}} \rangle \tag{1.27}$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{01}) = \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{01} \frac{\partial}{\partial \rho} n_{01} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{01}$$
$$- \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} + \gamma_0 \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) + \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_{\mathrm{i}}^{\mathrm{puff}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}} \tag{1.28}$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{02}) = \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{02} \frac{\partial}{\partial \rho} n_{02} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{02} + \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} \tag{1.29}$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{03}) = \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{03} \frac{\partial}{\partial \rho} n_{03} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{03} + f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \tag{1.30}$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' \bar{n}_z) = \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{nz} \frac{\partial}{\partial \rho} \bar{n}_z + \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \Delta D_{\mathrm{C}0} \frac{\partial}{\partial \rho} n_{\mathrm{C}0} - n_{\mathrm{e}} \bar{n}_z f_5 S_5 + n_{\mathrm{e}} n_{\mathrm{z}} A_6$$
$$+ \gamma_{0\mathrm{z}} \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + \gamma_{\mathrm{i,z}} \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) - (1 - \gamma_{0\mathrm{z}}) \nu_{\mathrm{L}} (\bar{n}_z - n_{\mathrm{C}0})$$
$$+ \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_{\mathrm{z}}^{\mathrm{puff}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}} \tag{1.31}$$

$$\frac{1}{V'} \frac{\partial}{\partial t} (V' g n_{\mathrm{b}}^{\mathrm{rp}}) = g \frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} + g \nu_{\mathrm{eff}} (\sqrt{\delta} n_{\mathrm{b}} - n_{\mathrm{b}}^{\mathrm{rp}}) - g \nu_{\mathrm{b}} n_{\mathrm{b}}^{\mathrm{rp}}$$

$$- \frac{1}{V'} \frac{\partial}{\partial \rho} V' g \left( \langle |\nabla \rho| \rangle u_{\mathrm{b}}^{\mathrm{rp}} n_{\mathrm{b}}^{\mathrm{rp}} - \langle |\nabla \rho|^2 \rangle D_{\mathrm{b}}^{\mathrm{rp}} \frac{\partial}{\partial \rho} n_{\mathrm{b}}^{\mathrm{rp}} \right) \tag{1.32}$$

## 2 Additional important relationship

$$\bar{n}_z = \sum_{i=0}^{\max(Z)-1} n_i \tag{2.1}$$

$$B_p = \frac{|\nabla \rho|}{R} \frac{\partial \psi}{\partial \rho}, \quad \langle B_p^2 \rangle = \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left( \frac{\partial \psi}{\partial V} \right)^2 \tag{2.2}$$

$$B_t = \frac{I}{R} = \frac{1}{R} \frac{4\pi^2}{\langle R^{-2} \rangle} \frac{\partial \psi_t}{\partial V}, \quad \langle B_t^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle} \left( \frac{\partial \psi_t}{\partial V} \right)^2 \tag{2.3}$$

$$\langle B^2 \rangle = \langle B_t^2 \rangle + \langle B_p^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle} \left( \frac{\partial \psi_t}{\partial V} \right)^2 + \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left( \frac{\partial \psi}{\partial V} \right)^2 \tag{2.4}$$

$$\langle B^\theta \rangle = \frac{4\pi^2}{V'} \frac{\partial \psi}{\partial \rho} = 4\pi^2 \frac{\partial \psi}{\partial V} = \frac{I \langle R^{-2} \rangle}{q} = 2\pi \left( \oint \frac{dl_p}{B_p} \right)^{-1} \tag{2.5}$$

$$\hat{q}^2 \equiv \frac{I^2}{2\langle B_p^2 \rangle} \left( \left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \tag{2.6}$$

$$\langle B^2 \rangle \langle R^2 \rangle - I^2 = \langle R^2 \rangle \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left( \frac{\partial \psi}{\partial V} \right)^2 (1 + 2\hat{q}^2) \tag{2.7}$$

$$\langle B \hat{q}_{s\|} \rangle \equiv \frac{2 \langle B q_{s\|} \rangle}{5 p_s} \tag{2.8}$$

$$\hat{u}_{s\theta} = \frac{\langle B u_{s\|} \rangle - B V_{1s}}{\langle B^2 \rangle} \tag{2.9}$$

$$\begin{aligned} \left\langle \frac{u_{s\zeta}}{R} \right\rangle &= \frac{1}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} \langle R u_{s\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} \langle B u_{s\|} \rangle \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + \frac{I}{\langle B^2 \rangle} \left( \left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) (\langle B u_{s\|} \rangle - B V_{1s}) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + I \left( \left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \hat{u}_{s\theta} \end{aligned} \tag{2.10}$$

$$\langle R \hat{q}_{s\zeta} \rangle = \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{2s} + \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{s\|} \rangle \tag{2.11}$$

$$B V_{1s} = -I \left( \frac{1}{e_s n_s} \frac{\partial n_s T_s}{\partial \psi} + \frac{\partial \Phi}{\partial \psi} \right), \quad B V_{2s} = -\frac{I}{e_s} \frac{\partial T_s}{\partial \psi} \tag{2.12}$$

$$u^\rho \equiv \langle \boldsymbol{u}\cdot\nabla\rho\rangle, \quad u^V = \langle \boldsymbol{u}\cdot\nabla V\rangle \tag{2.13}$$

$$u^\rho_g = -\frac{4\pi^2}{IV'\langle R^{-2}\rangle}\dot{\psi}_t, \quad u^V_g = -\frac{4\pi^2}{I\langle R^{-2}\rangle}\dot{\psi}_t = -\frac{V'}{2\rho\psi_{ta}}\dot{\psi}_t \tag{2.14}$$

$$\frac{1}{V'}\frac{\partial V'}{\partial t} = \frac{\partial}{\partial V}\langle \boldsymbol{u}_g\cdot\nabla V\rangle = \frac{\partial u^V_g}{\partial V} \tag{2.15}$$

$$\langle E_r\rangle = -\langle|\nabla\rho|^2\rangle\frac{\partial r}{\partial\rho}\frac{\partial}{\partial\rho}\Phi \tag{2.16}$$

$$\langle BE_\parallel\rangle = I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) \tag{2.17}$$

$$RE_\zeta = \dot{\psi} = \frac{V_{\rm loop}}{2\pi} \tag{2.18}$$

$$\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\rm i}\rangle = \hat{\mu}^{\rm i}_1(\langle Bu_{\rm i\parallel}\rangle - BV_{1\rm i}) + \hat{\mu}^{\rm i}_2\left(\frac{2\langle Bq_{\rm i\parallel}\rangle}{5p_{\rm i}} - BV_{2\rm i}\right) \tag{2.19}$$

$$\hat{\ell}^{ab}_{ij} \equiv \frac{\ell^{ab}_{ij}}{m_a n_a}, \quad \hat{\ell}^{\rm bb}_{ij} \equiv \frac{\ell^{\rm bb}_{ij}}{m_{\rm e}n_{\rm e}} \tag{2.20}$$

$$\Gamma^{\rm puff} \simeq \frac{\langle \boldsymbol{\Gamma}^{\rm puff}\cdot\nabla\rho\rangle}{\langle|\nabla\rho|\rangle} \tag{2.21}$$

## 3 Equations in the coordinates suitable for implementation

$$0 = \frac{\partial}{\partial V}\left[\langle|\nabla V|^2\rangle\frac{\partial}{\partial V}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_{\rm e} + Z_{\rm i}en_{\rm i} + Z_{\rm z}en_{\rm z} + Z_{\rm b}en_{\rm b} + Z_{\rm b}eg n^{\rm rp}_{\rm b}\right) \tag{3.1}$$

$$\begin{aligned}\epsilon_0\frac{\partial}{\partial t}\dot{\psi}_t = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\right] - \frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\right]\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\\ & + \frac{1}{\langle B^\theta\rangle}\left[-e\left(n_{\rm e}\langle Bu_{\rm e\parallel}\rangle - In_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle\right) + Z_{\rm i}e\left(n_{\rm i}\langle Bu_{\rm i\parallel}\rangle - In_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle\right)\right.\\ & \left. + Z_{\rm z}e\left(n_{\rm z}\langle Bu_{\rm z\parallel}\rangle - In_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle\right) + Z_{\rm b}e\left(n_{\rm b}\langle Bu_{\rm b\parallel}\rangle - In_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{3.2}$$

$$\begin{aligned}\frac{1}{c^2}\left\langle\frac{1}{R^2}\right\rangle\frac{\partial}{\partial t}\dot{\psi} = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial}{\partial V}\psi\right]\\ & - \mu_0\left(-en_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle + Z_{\rm i}en_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle + Z_{\rm z}en_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle + Z_{\rm b}en_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\end{aligned} \tag{3.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{3.4}$$

$$\frac{\partial}{\partial t}\frac{\psi_t}{\mu_0} = \frac{\dot{\psi}_t}{\mu_0} \tag{3.5}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e} = &-\frac{\partial}{\partial V}n_\mathrm{e}u_\mathrm{e}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{e} + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\ &+ Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e}-n_\mathrm{esup}) + (1-f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{3.6}$$

$$0 = -\frac{e}{m_\mathrm{e}}n_\mathrm{e}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}BV_{1\mathrm{e}} - \frac{e}{m_\mathrm{e}}In_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \frac{e}{m_\mathrm{e}}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle \tag{3.7}$$

$$\begin{aligned}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\|}\rangle = &-\frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}}\\ &+ \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{1}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}T_\mathrm{e} - \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{T_\mathrm{e}}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}n_\mathrm{e}\langle Bu_{\mathrm{i}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}n_\mathrm{e}\langle Bu_{\mathrm{z}\|}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle\\ &- \hat{\ell}_{12}^{\mathrm{ee}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ei}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ez}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{z}\|}\rangle\\ &- e\frac{n_\mathrm{e}}{m_\mathrm{e}}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) - \nu_\mathrm{L}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \frac{\partial}{\partial V}\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &- \frac{\partial}{\partial V}\left[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right.\\ &\left.-\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\partial}{\partial V}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right]\end{aligned} \tag{3.8}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle = &-(u_\mathrm{e}^V-u_g^V)\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial u_\mathrm{e}^V}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle]\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\langle Ru_{\mathrm{e}\zeta}\rangle\frac{\partial}{\partial V}n_\mathrm{e} - \frac{1}{m_\mathrm{e}}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}\frac{n_\mathrm{e}}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}\frac{n_\mathrm{e}}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}_{12}^{\mathrm{ee}} n_{\mathrm{e}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ee}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ei}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{ei}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- e \frac{n_{\mathrm{e}}}{m_{\mathrm{e}}} \dot{\psi} - e \frac{1}{m_{\mathrm{e}}} \frac{\partial \psi}{\partial V} n_{\mathrm{e}} u_{\mathrm{e}}^V \\
&- \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{e}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[ n_{\mathrm{e}} \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{e}} \frac{\partial}{\partial V} \Phi + \frac{C_{\mathrm{T}} - 1}{e} \left( \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{e}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle
\end{aligned} \tag{3.9}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{e}} T_{\mathrm{e}} = &- \frac{5}{2} (u_{\mathrm{e}}^V - u_g^V) \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{5}{2} \frac{\partial u_{\mathrm{e}}^V}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} [V_{\mathrm{e}}^{\mathrm{hp}} \langle |\nabla V| \rangle n_{\mathrm{e}} T_{\mathrm{e}}] \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- u_g^V \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - u_{\mathrm{i}}^V \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - u_{\mathrm{z}}^V \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle \\
&- \frac{3}{2} \nu_{\mathrm{Tei}} \left( n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{Tez}} \left( n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- 4\pi^2 \frac{\partial \psi}{\partial V} \dot{\psi}_t (-e n_{\mathrm{e}} \hat{u}_{\mathrm{e}\theta} + Z_{\mathrm{i}} e n_{\mathrm{i}} \hat{u}_{\mathrm{i}\theta} + Z_{\mathrm{z}} e n_{\mathrm{z}} \hat{u}_{\mathrm{z}\theta}) \\
&+ \dot{\psi} \left( -e n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + Z_{\mathrm{i}} e n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + Z_{\mathrm{z}} e n_{\mathrm{z}} \left\langle \frac{u_{\mathrm{z}\zeta}}{R} \right\rangle \right) \\
&- \nu_{\mathrm{L}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{esup}} T_{\mathrm{e}}) - \frac{3}{2} \nu_{\mathrm{LT_e}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} T_{\mathrm{esup}}) - E_{\mathrm{H}} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\
&+ f_{\mathrm{ei}} P_{\mathrm{b}} + P_{\alpha \mathrm{e}} + P_{\mathrm{RFe}} - P_{\mathrm{Br}}
\end{aligned} \tag{3.10}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{e}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle = &- \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \frac{1}{m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} B V_{1\mathrm{e}} \\
&+ \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{e}}}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- \hat{\ell}_{21}^{\mathrm{ee}} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ei}} n_{\mathrm{e}} \langle B u_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ez}} n_{\mathrm{e}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{eb}} \frac{n_{\mathrm{e}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ee}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ei}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ez}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned} \tag{3.11}
$$

$$
0 = - n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle \tag{3.12}
$$

$$0 = -n_{\mathrm{e}} B V_{1\mathrm{e}} + \frac{1}{e}\frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.13}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} = &-\frac{\partial}{\partial V} n_{\mathrm{i}} u_{\mathrm{i}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{i}} + \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) - \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) \\ &+ \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}\end{aligned} \tag{3.14}$$

$$0 = Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1\mathrm{i}} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} I n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \langle B^2 \rangle n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \tag{3.15}$$

$$\begin{aligned}n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B u_{\mathrm{i}\parallel} \rangle = &-\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} B V_{1\mathrm{i}} \\ &- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} n_{\mathrm{i}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} n_{\mathrm{i}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\ &- \hat{\ell}_{12}^{\mathrm{ie}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{ii}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{iz}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\ &+ Z_{\mathrm{i}} e \frac{n_{\mathrm{i}}}{m_{\mathrm{i}}} I \langle R^{-2} \rangle \left( -\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\ &- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{LC}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \left( \frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \left( \frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &- \frac{\partial}{\partial V} \left[ \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right. \\ &\left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right]\end{aligned} \tag{3.16}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle = &-(u_{\mathrm{i}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial u_{\mathrm{i}}^{V}}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \langle R u_{\mathrm{i}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} - \frac{1}{m_{\mathrm{i}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} \frac{n_{\mathrm{i}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\|} \rangle \\
&+ Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \dot{\psi} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \frac{\partial \psi}{\partial V} n_{\mathrm{i}} u^V_{\mathrm{i}} \\
&+ Z_{\mathrm{i}} \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{i}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[ (n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + (n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial}{\partial V} \Phi \right. \\
&\left. + \frac{C_{\mathrm{T}} - 1}{e} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left( \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{i}} + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{b}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle
\end{aligned} \tag{3.17}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{i}} T_{\mathrm{i}} = &- \frac{5}{2} (u^V_{\mathrm{i}} - u^V_g) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{5}{2} \frac{\partial u^V_{\mathrm{i}}}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} V^{\mathrm{hp}}_{\mathrm{i}} \langle |\nabla V| \rangle n_{\mathrm{i}} T_{\mathrm{i}} \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle T_{\mathrm{i}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&+ (u^V_{\mathrm{i}} - u^V_g) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} \nu_{\mathrm{Tei}} \left( \frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{e}} T_{\mathrm{e}} \right) - \frac{3}{2} \nu_{\mathrm{Tiz}} \left( n_{\mathrm{i}} T_{\mathrm{i}} - \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- \nu_{\mathrm{L}} \left( n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{isup}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{LT_i}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{1}{n_{\mathrm{b}}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \frac{\langle B u_{\mathrm{i}\|} \rangle}{\langle B^2 \rangle} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + P_{\alpha \mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{3.18}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{i}\|} \rangle = &- \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_2 \langle B u_{\mathrm{i}\|} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 \langle B \hat{q}_{\mathrm{i}\|} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_2 B V_{1\mathrm{i}} \\
&- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&- \hat{\ell}^{\mathrm{ie}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{e}\|} \rangle - \hat{\ell}^{\mathrm{ii}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{i}\|} \rangle - \hat{\ell}^{\mathrm{iz}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{z}\|} \rangle - \hat{\ell}^{\mathrm{ib}}_{21} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{ie}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\|} \rangle + \hat{\ell}^{\mathrm{ii}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle + \hat{\ell}^{\mathrm{iz}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\|} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle
\end{aligned} \tag{3.19}
$$

$$
0 = - n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{i}} \langle B u_{\mathrm{i}\|} \rangle \tag{3.20}
$$

$$0 = -n_{\mathrm{i}} B V_{1\mathrm{i}} - \frac{1}{Z_{\mathrm{i}} e} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.21}$$

$$\frac{\partial}{\partial t} n_{\mathrm{z}} = -\frac{\partial}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{z}} + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 - \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{isup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}} \tag{3.22}$$

$$0 = Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1\mathrm{z}} + Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} I n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} \langle B^2 \rangle n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \tag{3.23}$$

$$\begin{aligned}
n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel} \rangle = & -\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \frac{1}{m_{\mathrm{z}}} \hat{\mu}_1^{\mathrm{i}} B V_{1\mathrm{z}} \\
& - \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{z}}}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} \\
& + \hat{\ell}_{11}^{\mathrm{ze}} n_{\mathrm{z}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} n_{\mathrm{z}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
& - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
& + Z_{\mathrm{z}} e \frac{n_{\mathrm{z}}}{m_{\mathrm{z}}} I \langle R^{-2} \rangle \left( -\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\
& - \nu_{\mathrm{L}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
& - \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \left( \frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
& + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
& - \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \left( \frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{z}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
& - \frac{\partial}{\partial V} \left[ \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{z}} \right. \\
& \left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{z}} \right]
\end{aligned} \tag{3.24}$$

$$\begin{aligned}
\frac{\partial}{\partial t} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle = & -(u_{\mathrm{z}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \\
& + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \langle R u_{\mathrm{z}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{z}} - \frac{1}{m_{\mathrm{z}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
& + \hat{\ell}_{11}^{\mathrm{ze}} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} \frac{n_{\mathrm{z}}}{n_{\mathrm{i}}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle \\
& - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left( \langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned}$$

$$
\begin{aligned}
&+\hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{i}\|}\rangle \\
&+\hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}}\dot{\psi} + Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}}\frac{\partial \psi}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} \\
&+ Z_{\mathrm{z}}\frac{e\langle|\nabla V|^2\rangle D_{\mathrm{e}}}{m_{\mathrm{z}} T_{\mathrm{e}}}\frac{\partial \psi}{\partial V}\left[n_{\mathrm{z}}\frac{\partial \psi}{\partial V}\frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{z}}\frac{\partial}{\partial V}\Phi\right. \\
&\left.+\frac{C_{\mathrm{T}}-1}{e}\frac{n_{\mathrm{z}}}{n_{\mathrm{e}}}\left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}}\frac{\partial}{\partial V} n_{\mathrm{e}}\right) + \frac{T_{\mathrm{e}}}{e}\frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{z}}\right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{3.25}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{\partial}{\partial t} n_{\mathrm{z}} T_{\mathrm{z}} &= -\frac{5}{2}(u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{5}{2}\frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V} V_{\mathrm{z}}^{\mathrm{hp}}\langle|\nabla V|\rangle n_{\mathrm{z}} T_{\mathrm{z}} \\
&+ \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle T_{\mathrm{z}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&+ (u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta}\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle - \frac{3}{2}\nu_{\mathrm{Tez}}\left(\frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{e}} T_{\mathrm{e}}\right) - \frac{3}{2}\nu_{\mathrm{Tiz}}\left(\frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{i}} T_{\mathrm{i}}\right) \\
&- \nu_{\mathrm{L}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{zsup}} T_{\mathrm{z}}\right) - \frac{3}{2}\nu_{\mathrm{LT_z}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}} T_{\mathrm{zsup}}\right) + P_{\alpha\mathrm{z}} + P_{\mathrm{RFz}}
\end{aligned} \tag{3.26}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{z}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{z}\|}\rangle &= -\frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \frac{1}{m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle + \frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} \\
&- \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{1}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{T_{\mathrm{z}}}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&- \hat{\ell}_{21}^{\mathrm{ze}} n_{\mathrm{z}}\langle B u_{\mathrm{e}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zi}} n_{\mathrm{z}}\langle B u_{\mathrm{i}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zz}} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zb}}\frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}}\langle B u_{\mathrm{b}\|}\rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ze}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zi}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zz}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle
\end{aligned} \tag{3.27}
$$

$$
0 = -n_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle \tag{3.28}
$$

$$
0 = -n_{\mathrm{z}} B V_{1\mathrm{z}} - \frac{1}{Z_{\mathrm{z}} e}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V}\Phi \tag{3.29}
$$

$$\frac{\partial}{\partial t}n_\mathrm{b} = -\frac{\partial}{\partial V}n_\mathrm{b}u_\mathrm{b}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{b} + \frac{P_\mathrm{b}}{E_\mathrm{b}} - g\frac{P_{\mathrm{b}\perp}}{E_\mathrm{b}} - \nu_\mathrm{b}n_\mathrm{b} - \nu_\mathrm{bL}n_\mathrm{b} - \nu_\mathrm{eff}g(\sqrt{\delta}n_\mathrm{b} - n_\mathrm{b}^\mathrm{rp})$$
$$+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle f_t D_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} \tag{3.30}$$

$$0 = Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}n_\mathrm{b}BV_{1\mathrm{b}} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}In_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle - Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\langle B^2\rangle n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle \tag{3.31}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle = -\frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle + \frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}BV_{1\mathrm{b}}$$
$$+ \hat{\ell}_{11}^{\mathrm{bb}}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle + \sum_{a\neq\mathrm{b}}\hat{\ell}_{11}^{\mathrm{b}a}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}}\langle Bu_{a\parallel}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q} + \dot{\psi}\right) - \nu_\mathrm{bL}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle - \nu_{0\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle$$
$$- \nu_\mathrm{CX}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle + \frac{1}{m_\mathrm{b}}\langle BS_{\mathrm{m}\parallel\mathrm{b}}\rangle \tag{3.32}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle = u_g^V\frac{\partial}{\partial V}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{\partial}{\partial V}\langle Ru_{\mathrm{b}\zeta}\rangle n_\mathrm{b}u_\mathrm{b}^V$$
$$+ \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{be}}\frac{1}{n_\mathrm{e}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bi}}\frac{1}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bz}}\frac{1}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bb}}\frac{1}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}\dot{\psi} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\frac{\partial\psi}{\partial V}n_\mathrm{b}u_\mathrm{b}^V$$
$$- \nu_\mathrm{bL}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{CX}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\frac{I}{\langle B^2\rangle}\langle BS_{\mathrm{m}\parallel\mathrm{b}}\rangle \tag{3.33}$$

$$0 = -n_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I}n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle \tag{3.34}$$

$$0 = -n_\mathrm{b}BV_{1\mathrm{b}} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}In_\mathrm{b}\frac{\partial}{\partial V}T_\mathrm{b} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}IT_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} - n_\mathrm{b}\frac{\partial V}{\partial\psi}I\frac{\partial}{\partial V}\Phi \tag{3.35}$$

$$\frac{\partial}{\partial t}n_{01} = -\frac{\partial u_g^V}{\partial V}n_{01} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{01}\frac{\partial}{\partial V}n_{01} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{01}$$
$$- \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} + \gamma_0\nu_\mathrm{L}(n_\mathrm{i} - n_\mathrm{isup}) + \frac{\partial}{\partial V}V'\Gamma^\mathrm{puff}\Big|_\mathrm{b} \tag{3.36}$$

$$\frac{\partial}{\partial t}n_{02} = -\frac{\partial u_g^V}{\partial V}n_{02} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{02}\frac{\partial}{\partial V}n_{02} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{02} + \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} \tag{3.37}$$

$$\frac{\partial}{\partial t}n_{03} = -\frac{\partial u_g^V}{\partial V}n_{03} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{03}\frac{\partial}{\partial V}n_{03} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{03} + f_\mathrm{CX}\frac{P_\mathrm{b}}{E_\mathrm{b}} \tag{3.38}$$

$$\frac{\partial}{\partial t}\bar{n}_z = -\frac{\partial u_g^V}{\partial V}\bar{n}_z + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{nz}\frac{\partial}{\partial V}\bar{n}_z + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle \Delta D_{\mathrm{C0}}\frac{\partial}{\partial V}n_{\mathrm{C0}} - n_{\mathrm{e}}\bar{n}_z f_5 S_{\bar{5}} + n_{\mathrm{e}} n_{\mathrm{z}} A_6$$
$$+ \gamma_{0\mathrm{z}}\nu_{\mathrm{L}}\left(n_{\mathrm{z}} - n_{\mathrm{zsup}}\right) + \gamma_{\mathrm{i,z}}\nu_{\mathrm{L}}\left(n_{\mathrm{i}} - n_{\mathrm{isup}}\right) - (1-\gamma_{0\mathrm{z}})\nu_{\mathrm{L}}\left(\bar{n}_z - n_{\mathrm{C0}}\right)$$
$$+ \frac{\partial}{\partial V}V'\Gamma_{\mathrm{z}}^{\mathrm{puff}}\Big|_{\mathrm{b}} \tag{3.39}$$

$$\frac{\partial}{\partial t}(g n_{\mathrm{b}}^{\mathrm{rp}}) = g\frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} + g\nu_{\mathrm{eff}}(\sqrt{\delta}n_{\mathrm{b}} - n_{\mathrm{b}}^{\mathrm{rp}}) - g\nu_{\mathrm{b}} n_{\mathrm{b}}^{\mathrm{rp}}$$
$$- \frac{\partial}{\partial V}g\langle|\nabla V|\rangle u_{\mathrm{b}}^{\mathrm{rp}} n_{\mathrm{b}}^{\mathrm{rp}} + \frac{\partial}{\partial V}g\langle|\nabla V|^2\rangle D_{\mathrm{b}}^{\mathrm{rp}}\frac{\partial}{\partial V}n_{\mathrm{b}}^{\mathrm{rp}} \tag{3.40}$$

# 4 Boundary Conditions

| num. | name | variable | highest order derivative | center | edge |
|---|---|---|---|---|---|
| 1 | LQm1 | $\Phi$ | 2 | N | 0 |
| 2 | LQm2 | $\dot{\psi}_t$ | 0 | × | × |
| 3 | LQm3 | $\dot{\psi}$ | 0 | × | × |
| 4 | LQm4 | $\psi$ | 2 | N | $2\pi\mu_0 I_\mathrm{p}$ |
| 5 | LQm5 | $\psi_t/\mu_0$ | 2 | 0 | $\langle\lvert\nabla V\rvert^2/R^2\rangle\lvert_{V_b} I_\mathrm{vac}/(4\pi^2)$ |
| 6 | LQe1 | $n_\mathrm{e}$ | 2 | N | N† |
| 7 | LQe2 | $n_\mathrm{e}u_\mathrm{e}^V$ | 1 | 0 | × |
| 8 | LQe3 | $\langle Bu_{\mathrm{e}\parallel}\rangle$ | 2(0) | N(×) | N(×) |
| 9 | LQe4 | $n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle$ | 2 | N | N |
| 10 | LQe5 | $n_\mathrm{e}T_\mathrm{e}$ | 2 | N | N |
| 11 | LQe6 | $\langle B\hat{q}_{\mathrm{e}\parallel}\rangle$ | 0 | × | × |
| 12 | LQe7 | $n_\mathrm{e}\langle u_{\mathrm{e}\zeta}/R\rangle$ | 0 | × | × |
| 13 | LQe8 | $BV_{1\mathrm{e}}$ | 0 | × | × |
| 14 | LQi1 | $n_\mathrm{i}$ | 2 | N | N† |
| 15 | LQi2 | $n_\mathrm{i}u_\mathrm{i}^V$ | 1 | 0 | × |
| 16 | LQi3 | $\langle Bu_{\mathrm{i}\parallel}\rangle$ | 2(0) | N(×) | N(×) |
| 17 | LQi4 | $n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle$ | 2 | N | N |
| 18 | LQi5 | $n_\mathrm{i}T_\mathrm{i}$ | 2 | N | N |
| 19 | LQi6 | $\langle B\hat{q}_{\mathrm{i}\parallel}\rangle$ | 0 | × | × |
| 20 | LQi7 | $n_\mathrm{i}\langle u_{\mathrm{i}\zeta}/R\rangle$ | 0 | × | × |
| 21 | LQi8 | $BV_{1\mathrm{i}}$ | 0 | × | × |
| 22 | LQz1 | $n_\mathrm{z}$ | 2 | N | N† |
| 23 | LQz2 | $n_\mathrm{z}u_\mathrm{z}^V$ | 1 | 0 | × |
| 24 | LQz3 | $\langle Bu_{\mathrm{z}\parallel}\rangle$ | 2(0) | N(×) | N(×) |
| 25 | LQz4 | $n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle$ | 2 | N | N |
| 26 | LQz5 | $n_\mathrm{z}T_\mathrm{z}$ | 2 | N | N |
| 27 | LQz6 | $\langle B\hat{q}_{\mathrm{z}\parallel}\rangle$ | 0 | × | × |
| 28 | LQz7 | $n_\mathrm{z}\langle u_{\mathrm{z}\zeta}/R\rangle$ | 0 | × | × |
| 29 | LQz8 | $BV_{1\mathrm{z}}$ | 0 | × | × |
| 30 | LQb1 | $n_\mathrm{b}$ | 0(2) | ×(N) | ×(N) |
| 31 | LQb2 | $n_\mathrm{b}u_\mathrm{b}^V$ | 1 | 0 | 0‡ |
| 32 | LQb3 | $n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle$ | 0 | × | × |
| 33 | LQb4 | $n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$ | 0 | × | × |
| 34 | LQb7 | $n_\mathrm{b}\langle u_{\mathrm{b}\zeta}/R\rangle$ | 0 | × | × |
| 35 | LQb8 | $BV_{1\mathrm{b}}$ | 0 | × | × |
| 36 | LQn1 | $n_{01}$ | 2 | N | $\langle\lvert\nabla V\rvert\rangle S_\mathrm{gas}\rvert_b$ |
| 37 | LQn2 | $n_{02}$ | 2 | N | 0 |
| 38 | LQn3 | $n_{03}$ | 2 | N | 0 |
| 39 | LQnz | $\bar{n}_z$ | 2 | N | $\langle\lvert\nabla V\rvert\rangle S_{\mathrm{gas,z}}\rvert_b$ |
| 40 | LQr1 | $n_\mathrm{b}^\mathrm{rp}$ | 0(2) | ×(N) | ×(N) |

0: Dirichlet condition ($u = 0$), N: Neumann condition ($u' = 0$)
value: Neumann condition ($u' =$ value), ×: no condition
†: imposed as a constraint in V5.60 (`MDLWBC` = 1), see Sec. 7.2

NOTE: The dependent variable of LQm4 turns to be $\psi/\mu_0$ when NB is applied. Subsequently, the edge boundary condition listed for that variable is also changed to $2\pi I_\mathrm{p}$.

NOTE: Rows 2 to 5 deserve a word, because the variable that carries the second-order operator is not the variable of the equation in which the operator is written. The LQm2 equation advances $\dot{\psi}_t$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \langle R^{-2}\rangle^{-1} \frac{\partial}{\partial V}(\psi_t/\mu_0)]$; the LQm3 equation advances $\dot{\psi}$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \frac{\partial}{\partial V}\psi]$. Counting per dependent variable over the whole system, as this section does, $\dot{\psi}_t$ and $\dot{\psi}$ are never differentiated and so take order 0 and no condition of their own, while $\psi_t/\mu_0$ and $\psi$ take order 2 and carry the two conditions each. The table above now reads that way. In the implementation the two conditions of each pair are applied to the LQm2 and LQm3 equation rows, since that is where the boundary term left by the integration by parts appears; this is the correct weak form and is not a discrepancy with the table. The axis condition of $\psi_t/\mu_0$ is imposed there as $\dot{\psi}_t|_{V=0} = 0$, which holds $\psi_t(0)$ at its initial value, namely zero, and is therefore the Dirichlet condition listed.

NOTE: In the table the column labeled by the highest order derivative means the order of the highest order derivative of a dependent variable, not an equation, in the corresponding row. Please note that the term not integrated by parts is never counted. For example, the first row shows the highest order of derivatives is equal to 2 for $\Phi$ because the LQm1 equation has the diffusion term having the second derivative of $\Phi$ with respect to $V$. In addition, the LQe2 and LQi2 equations also have the first derivative of $\Phi$, which is not a dependent variable of these equations. As a whole, the highest order derivative of $\Phi$ in this equation system is 2 that comes from the LQm1 equation. When we look at $n_\mathrm{e}$, we find that the highest order derivative of $n_\mathrm{e}$ is 2 shown in the LQe5 equation, even though no derivative appears in the LQe1 equation. The Neumann conditions for $n_\mathrm{e}$ come from the LQe5 equation. When it comes to $n_\mathrm{e}u_\mathrm{e}^V$, the highest derivative is found as order 1 in the LQe1 equation, but this term is not integrated by parts. It means that any boundary condition is not naturally imposed on $n_\mathrm{e}u_\mathrm{e}^V$. The Dirichlet boundary condition should be imposed at axis.

# 5 Momentum pinch of V5.62

V5.62 changes the turbulent momentum pinch of the thermal species. The flux of toroidal angular momentum of species $s$ is written with the density outside the gradient (the form of the scaling used below),

$$\Gamma_{\mathrm{L},s} = m_s \left( \langle |\nabla V| \rangle \, V_\mathrm{p} \, n_s \langle R u_{s\zeta} \rangle - \langle |\nabla V|^2 \rangle \, \mu_s \, n_s \, \partial_V \langle R u_{s\zeta} \rangle \right), \tag{5.1}$$

with the momentum diffusivity $\mu_s$ and the pinch velocity $V_\mathrm{p}$ ($V_\mathrm{p} > 0$ outward). The pinch is

$$V_\mathrm{p} = -\frac{\mu_\mathrm{i}}{R_0} \left[ 1.1 \, g\!\left( \frac{R_0}{L_n} \right) + 1.0 \right] T(\rho), \qquad \frac{R_0}{L_n} = -R_0 \, \frac{d \ln n_\mathrm{e}}{d\rho_V}, \tag{5.2}$$

where $\rho_V$ is the volume radius and $n_\mathrm{e}$ is smoothed as before.

**Origin and range of the scaling.** The multi-machine scaling of Tala et al. [4] is $-R_0 V_\mathrm{pinch}/\chi_\phi = 1.1 \, R/L_n + 1.0$, both terms inward. It was obtained from NBI modulation in JET, DIII-D and AUG and from magnetic perturbations in NSTX, with the pinch and the Prandtl number averaged over $0.4 < \rho_\mathrm{tor} < 0.8$; the data cover $R/L_n$ = 1.2–3.6 (0.2–3.8 with NSTX) and pinch numbers 0.9–6.2. The slopes of the single devices are 1.2 (JET, DIII-D), 1.5 (AUG) and 0.6 (NSTX); gyrokinetic calculations give a dependence about half as strong. The profiles are "not accurate in the edge region", and an edge torque added to the analysis does not change the coefficients inside $r/a < 0.8$. The analysis method [6] and the JET subset [5] write the flux in the form above; residual stress and intrinsic torque are neglected. No upper limit of $R/L_n$ is stated. Beyond the range, the JETPEAK database [7] ($R/L_n$ = 1–10, $r/a$ up to 0.85) gives an $R/L_n$ coefficient of 0.41–0.66 for the steady-state rotation gradient (pinch and residual stress together), and the regression of GKW calculations 0.44; neither shows a saturation.

**The closure.** Equation (5.2) is a reduced closure, not the Tala law extrapolated over the whole plasma: it is the Tala fit inside its data, it takes the slope beyond the data from the multi-machine and gyrokinetic information, and it is not applied in the edge and the SOL.

- $g(x) = x$ for $|x| \le x_1$ and

$$g(x) = \mathrm{sgn}(x) \left[ x_1 + s(|x| - x_1) + (1 - s) \, w \tanh \frac{|x| - x_1}{w} \right] \quad (|x| > x_1), \tag{5.3}$$

  with $x_1 = 3.8$ (`PCHRLN`, the largest $R/L_n$ of the fit), $s = 0.4$ (`PCHSLP`, so that the slope goes from 1.1 to 0.44) and $w = 1$ (`PCHWID`). $g$ is continuous up to its second derivative. The pinch number is 7.7 at $R/L_n$ = 8 and 8.6 at 10, within the measured pinch numbers (up to about 8), against 9.8 and 12 for the linear law. $s = 0$ saturates the pinch number at $1 + 1.1(x_1 + w)$; this is an experimental option without a basis in the literature.

- $T(\rho) = 1$ for $\rho \le \rho_1 = 0.8$ (`RhoPch1`, the outer edge of the data), 0 for $\rho \ge \rho_2 = 0.9$ (`RhoPch2`) and the quintic smoothstep $1 - \xi^3(10 - 15\xi + 6\xi^2)$, $\xi = (\rho - \rho_1)/(\rho_2 - \rho_1)$, between. $T$ is continuous up to its second derivative and exactly zero outside $\rho_2$, so that the pinch carries no momentum through the wall. The location follows the data (0.8), the momentum boundary of the TOPICS validation (0.9 [8]) and the pedestal-top boundaries of present integrated modelling (0.8–0.94); the width is a numerical choice. Outside $\rho_2$ the momentum is transported by the diffusion only. Cubic, cosine and tanh tapers with the same end points give the same rotation to three digits.

- $V_\mathrm{p}$ is built from the ion viscosity and given to every thermal species; the beam has no turbulent momentum flux. With a common $\mu$ the fluxes sum to $V_\mathrm{p}L - \mu\sum_s m_s n_s \partial_V \langle R u_{s\zeta}\rangle$ times the metric factors, the pinch of the total thermal momentum $L = \sum_s m_s n_s \langle R u_{s\zeta}\rangle$ used in TOPICS [8], whether or not the species rotate alike. With species-dependent `rMus0` the pinch number $-R_0 V_\mathrm{p}/\mu_s$ differs between species and this equivalence is lost.

**Correction of the sign.** Up to V5.61 the code evaluated $V_\mathrm{p} = (\mu/R_0)(1.1R_0\, d\ln n_\mathrm{e}/d\rho_V + 1)$, i.e. $-R_0V_\mathrm{p}/\mu = 1.1R_0/L_n - 1$: the constant term pointed outward, against the scaling it cited. V5.62 uses $+1.0$ (verified on the output pinch velocity to three digits). The commented-out edge suppressions of the old code (a tanh at $\rho = 0.98$ and a Gaussian at $\rho = 1$) had no documented basis and are removed. Without the radial restriction the old pinch reached a pinch number of 14 at $\rho = 0.9$ and about 140 at the separatrix, and the convective flux at the wall carried momentum into the plasma (+0.04 N m in the JT-60U case).

**Validation.** JT-60U-like case of Sec. 6.7 (3.2 MW co-tangential and 0.4 MW perpendicular NBI, $\chi = \mu = 0.22(1 + 19\rho^2)$ m$^2$/s, `PNsSUP(1)` $= 0.0125$ as explained below), production defaults (`FSTPTM` $= 3$), 1.5 s:

| case | $u_{\mathrm{i}\phi}(0)$ km/s | $u_{\mathrm{i}\phi}(0.82)$ km/s | $E_r(0.5)$ kV/m | $j_r(0.97)$ A/m$^2$ | $j \times B$ torque N m | $L_\phi$ N m s |
|---|---|---|---|---|---|---|
| no ripple | 510 | 57 | +44 | −0.017 | +0.16 | 2.03 |
| half ripple (FST) | 370 | 27 | +22 | −0.103 | −3.25 | 1.58 |
| full ripple | 198 | −8 | −5 | −0.180 | −7.28 | 1.05 |

The line-averaged density is $1.15\times10^{19}$ m$^{-3}$ and the thermal mass inside $\rho = 1$ is 1.54–1.62 mg. The first version of this table used `PNsSUP(1)` $= 0.01$ (the default) and `FSTPTM` $= 2$: 621, 441 and 226 km/s ($\bar n_\mathrm{e} = 1.02\times10^{19}$ m$^{-3}$, thermal mass 1.29 mg); `FSTPTM` $= 3$ gives 620 and 223 km/s without and with full ripple at 0.01, and V5.61 gave 658, 417 and 128 km/s. The ripple changes the rotation, $E_r$, the return current and its torque at `PNsSUP(1)` $= 0.0125$ as at 0.01. The rest of this paragraph refers to `PNsSUP(1)` $= 0.01$. The pinch number is 6.3 at $\rho = 0.5$ and 7.9 at 0.8 and vanishes outside 0.9; in V5.61 it was 14 at 0.9. The profile is more peaked in the core (the constant term) and lower at the edge ($u_{\mathrm{i}\phi}(0.9) = 46$ against 105 km/s). The volume integrals close to $2\times10^{-5}$ of the injected torque in every row, the pinch flux at the wall is zero and the friction terms sum to $3\times10^{-9}$ N m. With `NRMAX` $= 120$ the total angular momentum agrees with `NRMAX` $= 60$ to 0.5% up to 0.8 s (the Picard iteration is limited there by the known impurity terms at the axis, as in V5.61). On the test equilibria of Sec. 6.7 densities and temperatures change by less than $3\times10^{-3}$ and the rotation by 13–40%; the mean Picard count rises by 1–4 and no step more reaches `ICMAX`. The Tala law with the corrected sign but without the radial restriction and the slope reduction spins the JT-60U case up to 881 km/s and stops at 1.4 s at the impurity grid mode of supersonic rotation.

**Density of the benchmark.** `PNsSUP(1)` is the upstream SOL density and sets the density level of the whole plasma (Sec. 7.8); the shape of the density profile does not depend on it. The benchmark input sets it to 0.0125 ($10^{20}$ m$^{-3}$), which gives $\bar n_\mathrm{e} = 1.15\times10^{19}$ m$^{-3}$ and $n_\mathrm{e}(\rho = 1) = 0.22\times10^{19}$ m$^{-3}$. The line-averaged density is the one estimated from the measured profile of the JT-60U L-mode discharge compared in [1] (Fig. 1 of [2], which gives no value), and the separatrix density lies at the lower end of the range expected for JT-60U L-modes (0.2–0.45$\times10^{19}$ m$^{-3}$). This value normalizes the density of this benchmark; it is not a universal value, and the default of `PNsSUP(1)` stays 0.01. A new device, discharge, equilibrium, heating scheme or particle-transport

model needs its own value, found by an outer loop of two or three runs to the target density (secant in log–log; there is no density feedback inside the code). The benchmark is JT-60U-like and not the discharge: $I_\mathrm{p} = 1.0$ MA against 1.23 MA, a circular cross-section against a D-shaped plasma of 77 m$^3$, and the computed density profile falls more steeply towards the edge ($n_\mathrm{e}(0.9)/\bar{n}_\mathrm{e} = 0.45$ against 0.74). The rotation was not used to choose `PNsSUP(1)`: it is a consistency check of the momentum model, not a fitting target.

The reduced ripple model is judged by the changes with the ripple rather than by the absolute rotation. The co-rotation without ripple is 1.7 times that of [1] (510 against 300 km/s); the states of that work, restored from its save files with the 2008 code, show why. The 2008 plasma was pure deuterium with a flat density ($n_\mathrm{e} = 1.4$–$1.5\times10^{19}$ m$^{-3}$ up to $r/a = 0.7$), 1.4 times heavier than the present one (thermal mass 2.30 against 1.59 mg, moment of inertia 2.70 against $1.87 \times 10^{-5}$ kg m$^2$); its torque was 7.63 N m (85 keV, the whole beam velocity toroidal at $R_0$) with half of it deposited off axis, it had no momentum pinch, and its wall held $u_\phi = 0$. The momentum confinement times $L_\phi/T_\mathrm{NBI}$ of the thermal species are similar (0.137 and 0.143 s), so the mass-weighted rotation $R_0 L_\phi/I_\phi$ differs by 1.3 (132 and 174 km/s) and the more peaked profile gives the rest of the factor 1.7 on axis.

**Limitations.** The pinch number is taken from the electron density gradient alone; the dependences on $q$, $\hat{s}$, $\varepsilon$, $T_\mathrm{i}/T_\mathrm{e}$ found in the regressions are not included, and no residual stress is modelled. The radial restriction is fixed in $\rho$ and does not follow the pedestal of an H-mode. The momentum diffusivity itself is prescribed ($\mu = \chi$ in the benchmark). A Coriolis-type alternative with a geometric dependence ($-R_0 V/\mu \propto \sqrt{r/R}$, as used for DTT [9]) rotates the JT-60U case 30% more slowly; its coefficient is a calibration, not a universal constant, and it is not implemented in the production code.

## 6 Beam and ripple model of V5.61

V5.61 changes the beam and ripple equations. The model itself (its assumptions, the ordering behind each term and the comparison with the 2008 ripple model [1]) is described in `beam_ripple.tex`; this section states what the code solves and what changed.

| change | effect |
|---|---|
| ripple momentum terms of the beam removed (B-full) | `FSRP` = 1 with NBI |
| LQr1 written as the conservation law of $g n_\mathrm{b}^\mathrm{rp}$ | `FSRP` = 1 with NBI |
| $\Gamma_\mathrm{b}(\rho_b) = 0$ | `MDBEAM` = 1 with NBI |
| fast-ion friction $\ell_{11}^{\mathrm{f}k} = \ell_{11}^{k\mathrm{f}}$ restored | NBI (`MDLNEO` = 1, 2) |
| $Z_\mathrm{eff}$ and the fast-ion inventory weighted by $g$ | `FSRP` = 1 with NBI |
| `FSRP` restricted to 0, 1, 2; `FSRP` = 2 keeps $gP_{\mathrm{b}\perp}$ in $n_\mathrm{b}$ | input |
| on-axis ripple trapping rate from the interior | `FSRP` = 1 with NBI |

Runs without NBI are bit-identical to V5.60 (X-point and circular test equilibria, `NRMAX` = 60 and 240, `MDLNZD` = 0, `MDLNOFL` = 2, `FSRP` = 0 and 1). No default switch changes. For runs with NBI, and in particular with `FSRP` = 1, `NANDER` = −20 is recommended (Sec. 7.5): the impurity radial flux at the wall converges slowly without it once the beam heats the edge.

### 6.1 `FSRP` and the ripple-well fraction

`FSRP` is a mode, not a factor: 0, no ripple; 1, full ripple model (ripple-trapped ions, LQr1, and ripple diffusion of $n_\mathrm{b}$); 2, ripple diffusion of $n_\mathrm{b}$ without ripple-trapped ions. Other values are rejected. The ripple amplitude is $\delta(r,\theta) = \delta_\mathrm{min} I_0(N a_\delta / R_{\delta\mathrm{min}})$ with `DltRPn` $= \delta_\mathrm{min}$; `FSRP` only switches it on. $R_{\delta\mathrm{min}}$ (input `RDltRPn`, default 2.4 m) is the major radius on the midplane where the ripple amplitude has its minimum $\delta_\mathrm{min}$; the formula takes the surfaces $\delta$ = const as circles centred there, a property of the TF coil geometry (the coil shape and radial position), not of the plasma. 2.4 m is the JT-60U value that entered the 2008 code together with the ripple model ("specific value for JT-60U", also used by the routine that drew the ripple contours); it was fixed in the code up to V5.61.00, and other devices need their own value. The text of [1] gives 2.8 m, but its own numbers require 2.4 m: with $\delta_\mathrm{min} = 2\times10^{-6}$ and $N = 18$ the formula gives $\delta_a = 1.28\%$ at the outboard midplane and 0.30% at $\theta = \pi/2$ with 2.4 m, the values quoted in [1], and 0.085% and 0.043% with 2.8 m; the contours of its Fig. 1 are centred near 2.4 m. The 2.8 m of the text is a misprint. A point of a flux surface lies in a ripple well when

$$\alpha \equiv \frac{\epsilon|\sin\theta|}{N q \delta(r,\theta)} < 1, \tag{6.1}$$

and $g(\rho)$ (`rip_rat`) is the fraction of the poloidal circumference where it holds, found by searching the rims on the low- and high-field sides. $g$ is not zero near the axis: close to the low-field-side midplane $\sin\theta \to 0$ and a narrow arc always satisfies $\alpha < 1$ (for the JT-60U case below $g = 0.22$, 0.06 and 0.43 at $\rho = 0.04$, 0.25 and 0.96). V5.61 counts these narrow core wells as the criterion gives them; no cut-off is applied.

### 6.2 Ripple-trapped beam density: conservation form

$n_\mathrm{b}^\mathrm{rp}$ (LQr1) is the density of the ripple-trapped beam ions inside the wells; the inventory per unit volume is $N_\mathrm{r} = g n_\mathrm{b}^\mathrm{rp}$. V5.61 solves the conservation law of $N_\mathrm{r}$ with $n_\mathrm{b}^\mathrm{rp}$ as the unknown,

$$\frac{\partial}{\partial t}(g n_\mathrm{b}^\mathrm{rp}) + \frac{\partial}{\partial V}\left(g U n_\mathrm{b}^\mathrm{rp}\right) - \frac{\partial}{\partial V}\left(g K \frac{\partial}{\partial V} n_\mathrm{b}^\mathrm{rp}\right) = g\frac{P_{\mathrm{b}\perp}}{E_\mathrm{b}} + g\nu_\mathrm{eff}(\sqrt{\delta}\, n_\mathrm{b} - n_\mathrm{b}^\mathrm{rp}) - g\nu_\mathrm{b} n_\mathrm{b}^\mathrm{rp}, \tag{6.2}$$

$$U = \langle|\nabla V|\rangle u_\mathrm{b}^\mathrm{rp}, \qquad K = \langle|\nabla V|^2\rangle D_\mathrm{b}^\mathrm{rp}. \tag{6.3}$$

$g$ stays inside every divergence. Written for $N_\mathrm{r}$, the diffusive flux is $-gK\partial_V(N_\mathrm{r}/g) = -K\partial_V N_\mathrm{r} + K N_\mathrm{r}\partial_V \ln g$; the second part, the $dg/dV$ term, is what makes the flux vanish where the wells end ($g \to 0$ with $n_\mathrm{b}^\mathrm{rp}$ finite), so it cannot be dropped: without it the ripple-trapped ions diffuse out of the well region (a test with $-K\partial_V N_\mathrm{r}$ alone lost 80% of the inventory through the core-side rim). The form (6.2) contains no $1/g$.

Changes from V5.60, where LQr1 was $\partial_t n_\mathrm{b}^\mathrm{rp} + \partial_V(r u_\mathrm{b}^\mathrm{rp} n_\mathrm{b}^\mathrm{rp}) - \partial_V(K\partial_V n_\mathrm{b}^\mathrm{rp}) = \ldots$ without $g$:

- the $dg/dV$ terms (above) were missing and $n_\mathrm{b} + g n_\mathrm{b}^\mathrm{rp}$ was not conserved: the particle inventory of $n_\mathrm{b} + g n_\mathrm{b}^\mathrm{rp}$ was off by 2–9% of the NBI source;
- the advection used $r u_\mathrm{b}^\mathrm{rp}$ (a remnant of the $\psi = r^2$ coordinate of the earlier code) instead of $\langle|\nabla V|\rangle u_\mathrm{b}^\mathrm{rp}$, a factor $4\pi^2 R$ too weak in circular geometry;
- the source had no $g$, so $n_\mathrm{b}^\mathrm{rp}$ grew where there are no wells.

Discretization: the mass, source, exchange and thermalization terms are the element matrices $\int g\,(\ldots)\,w\,dV$; the exchange terms are the same matrices as in LQb1 with opposite sign, so $n_\mathrm{b} + g n_\mathrm{b}^\mathrm{rp}$ is conserved exactly by the discrete equations. The advection is in strong form, $-\int \partial_V(g U n_\mathrm{b}^\mathrm{rp}) w\,dV$; its sum over the test functions is

the wall flux $[gUn_{\rm b}^{\rm rp}]_{V_b}$, the ripple loss out of the plasma, and no boundary condition is needed at either end ($gU = 0$ at the axis). The diffusion has the natural condition. A node with no well on either side gets the identity $\partial_t n_{\rm b}^{\rm rp} = 0$. No SUPG: the element Péclet number is 15–30 at the edge and $10^2$–$10^3$ inside, but the solution shows no grid-scale structure and converges with the mesh (Sec. 6.7).

Every other equation already weighted $n_{\rm b}^{\rm rp}$ by $g$ (LQb1, Poisson, LQi1); V5.61 also weights $Z_{\rm eff}$ and the fast-ion inventory (`ANFINT`) by $g$.

### 6.3 Ripple momentum closure of the beam (B-full)

In V5.60 the parallel beam momentum (LQb3) had the two ripple terms of the 2008 model [1],

$$-\nu_{\rm eff} g\sqrt{\delta}\, m_{\rm b} n_{\rm b} \langle B u_{\rm b\parallel}\rangle, \qquad +\frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\langle B u_{\rm b\parallel}\rangle\, f_t D_{\rm b}\, \frac{\partial}{\partial V} n_{\rm b}, \tag{6.4}$$

the momentum removed by ripple trapping and the momentum carried by the ripple diffusion flux; the toroidal momentum (LQb4) had neither. V5.61 removes both, and adds nothing to LQb4. The reason is the order of these terms in the velocity moments of the beam distribution, not the numerical instability that the second term caused on linear elements:

- The ions scattered into the ripple wells have $|u_\parallel|/v < \sqrt{\delta}$. The momentum they take is $\int_{|\xi|<\sqrt{\delta}} mv\xi\, \nu f\, d^3v$, bounded by $\sqrt{\delta}\, mv$ per ion and $O(\delta)\, mv$ for a smooth $f(\xi)$, against $mu_{\rm b\parallel} = O(1)\, mv$ per ion in the 2008 form. The same holds for $mRv_\zeta$ up to the drift, $O(\rho_{\rm b}^*)$.

- Ripple diffusion acts on the banana ions. Their bounce-averaged distribution is even in $\xi$, so the parallel momentum flux $\int mv\xi\, \Gamma_r[f]\, d^3v$ vanishes at zero orbit width; the first non-zero part comes from the offset of the two legs of the banana, $O(\rho_{\rm b}^*)$. For the toroidal momentum, $\langle mRv_\zeta\rangle_{\rm b} = P_\zeta + Ze\langle\psi\rangle_{\rm b}$ with $v_\parallel = 0$ at the tips and $\langle v_\parallel\rangle_{\rm b} = 0$, again $O(\rho_{\rm b}^*)$.

- The beam fluid of TX keeps no finite-orbit-width physics (the beam viscosity is zero), so keeping these $O(\delta)$ and $O(\rho_{\rm b}^*)$ terms alone would be inconsistent. $\rho_{\rm b}^* = 0.1$–$0.2$ at the edge of JT-60U for 80 keV D: the terms are dropped because the closure does not retain that order, not because they are negligible.

Both components vanish, so the closure also respects the radial force balance (LQb2): a ripple term in LQb3 alone forces a spurious radial beam flux through the multiplier $\Gamma_{\rm b}$. Ripple particle transport is kept: the trapping, detrapping and diffusion of $n_{\rm b}$ in LQb1, and LQr1. The main effect of the ripple in the 2008 work, the torque of the return current, comes from the charge that these particle fluxes move and is unchanged (Sec. 6.7).

### 6.4 Radial beam flux at the wall

With `MDBEAM` = 1, $\Gamma_{\rm b} = n_{\rm b} u_{\rm b}^V$ (LQb2) is the Lagrange multiplier of the beam radial force balance; V5.60 had no condition on it at $\rho_b$. V5.61 imposes

$$\Gamma_{\rm b}(\rho_b) = 0. \tag{6.5}$$

This is a closure of the reduced model, not a statement that fast ions are not lost to the wall. The model has no surface or orbit-loss channel for the beam; outside the separatrix the beam is lost by the volume loss $\nu_{\rm bL} n_{\rm b}$. The wall value of the multiplier was not an outflow law: it was whatever the force balance at the wall node required, with the diamagnetic term of a beam with fixed $T_{\rm b} = 2E_{\rm b}/3$, whereas the fast-ion orbit width (10–20 cm) exceeds the gradient lengths of the SOL. Left free, that value pushed $n_{\rm b}$ at the wall node towards zero and

made the Picard iteration cycle without converging (a period-5 cycle of $\Gamma_\mathrm{b}$ on the circular test equilibrium). The alternatives were rejected: the SOL beam removal used with `MDBEAM` = 0, 2 rewrites $n_\mathrm{b}$ outside the equations and contradicts the constraint rows; an outflow $\Gamma_\mathrm{b} = n_\mathrm{b} v_\mathrm{out}$ needs a physical $v_\mathrm{out}$ and would double-count $\nu_\mathrm{bL}$. The condition changes runs with NBI by 1.5–5% in the core rotation for perpendicular injection and by 0.5% for the JT-60U case below. It should be revisited if a finite-orbit-width or orbit-loss model is introduced.

### 6.5 Beam–thermal friction

The friction between the beam and the thermal species follows the fast-ion coefficients of Kikuchi and Azumi [3] (`imodel_neo(3)` = 2, isotropic fast ions in the pitch-angle-scattering regime): $\ell_{11}^{\mathrm{fe}} = \ell_{11}^{\mathrm{ef}}$, $\ell_{11}^{\mathrm{f}k} = \ell_{11}^{k\mathrm{f}}$ ($k$ = i, z), $\ell_{11}^{\mathrm{ff}} = -\sum_k \ell_{11}^{\mathrm{f}k}$, the Maxwellian $\ell_{11}^{kk}$ reduced by $\ell_{11}^{k\mathrm{f}}$, and the heat friction $\ell_{21}^{\mathrm{ef}} = -\frac{3}{2}\ell_{11}^{\mathrm{ef}}$ with $\ell_{21}^{\mathrm{fe}} = 0$. The 11 block is symmetric, so the friction forces sum to zero; only the heat friction is not symmetric, as the paper states. In V5.60 the fast-ion row was obtained from the thermal row with the Braginskii ratio $\tau_{ee}/\tau_{ii}$, which assumes equal Coulomb logarithms, and converted back to physical units with the actual $\tau_{ee}$ and $\tau_{ii}$. The result was $\ell_{11}^{\mathrm{fi}}/\ell_{11}^{\mathrm{if}} = \ln\Lambda_{ee}/\ln\Lambda_{ii}$ (0.82–0.86 for D, 1.00–1.08 for C) and a net force of 2–10% of the fast-ion friction: 3% of the NBI torque in the JT-60U case, 13% at lower $T_\mathrm{e}$. V5.61 converts with the same collision times; the column sums of the 11 block are now $10^{-16}$ of the diagonal. Applicability: the coefficients assume an isotropic fast-ion distribution and the slowing-down regime $v_\mathrm{Ti} \ll v_\mathrm{b}$, and they ignore the fast-ion heat flow; for a tangential beam the distribution is anisotropic.

### 6.6 Discrete momentum conservation of the friction

The friction force on species $a$ is $\sum_b \ell^{ab} u_b$ with $\sum_a \ell^{ab} = 0$, so the forces cancel for any flows. The coefficients satisfy this at every node (to $10^{-8}$ of the diagonal, MI). The equations of species $a$ are written per unit mass density, so the term of row $a$ acting on the momentum $X_b = n_b\langle R u_b\rangle$ has the coefficient $\ell^{ab}/(m_a n_a) \times n_a/n_b$; up to V5.61.00 it was integrated as the product of the two interpolants (`fem_int(20,lab,Ratden)`, and `fem_int(20,lab,n)` in the parallel equations), while the diagonal term of the same column used the interpolated coefficient (`fem_int(2,lab)`). Per element the two differ by $-h\,\Delta a\,\Delta b\,(X_1 + X_2)/12$, where $\Delta a$, $\Delta b$ are the changes of the two factors across the element, so the columns did not sum to zero and the friction left a net force of $O(h^2)$. It was measured directly: the residual falls by 2.2–2.3 from `NRMAX` = 60 to 90 and by 3.8–3.9 to 120, and in the JT-60U case it came almost entirely from the ion–impurity pair near $\rho = 0.8$, where $n_i/n_z$ and the coefficients vary most. Every friction term of a column is now assembled from the nodal product of its row-dependent factors ($\ell^{ab} n_a/n_b$, $\ell^{ab} n_a$, and $\ell_{12}^{ab} n_a$ for the heat friction, with the row-independent factors $I/\langle B^2\rangle$ and $\partial_V T_b$ integrated alike in every row). The element matrices then sum to zero over a column whenever the nodal coefficients do, and the continuous equations are unchanged. The change of the solution is $O(h^2)$: in the JT-60U case at 1.5 s, $u_{\mathrm{i}\phi}(0)$ goes from 658 to 656 km/s, the total angular momentum by −0.2%, the impurity density by up to 1% and $E_r$ by up to 0.7% (normalised to their maxima); without NBI, where the spurious force was comparable to the whole toroidal angular momentum, the small rotation changes by up to 5%. The Picard counts are unchanged.

### 6.7 Validation

**Regression.** Without NBI all runs are bit-identical to V5.60 (see the start of this section). With NBI and `FSRP` = 0 the solution changes by the intended $\Gamma_\mathrm{b}(\rho_b) = 0$ and the friction correction: on the X-point test

equilibrium with 5 MW perpendicular NBI the total toroidal angular momentum at 0.5 s is 16% lower (the injected torque is zero there, so the spurious friction source was comparatively large), with 3 MW co-tangential plus 1 MW perpendicular 2% lower; bulk densities and temperatures change by less than $10^{-3}$. On the circular test equilibrium the wall cycle of $\Gamma_\mathrm{b}$ is gone (V5.60: 25 of 5000 steps at `ICMAX`; V5.61: none).

**Stability matrix.** `FSRP` = 1, perpendicular NBI, `NANDER` = −20, X-point and circular, 0.3 s (0.15 s for $\Delta t = 2.5 \times 10^{-5}$):

| case | X-point: steps at `ICMAX` / mean | circular |
|---|---|---|
| NRMAX 60, $\Delta t = 10^{-4}$, 2.5 / 5 / 10 MW | 1 / 15, 0 / 21, 0 / 28 | 0 / 19, 0 / 28, 0 / 34 |
| $\Delta t = 2.5 \times 10^{-5}$, $2 \times 10^{-4}$ | 0 / 11, 0 / 28 | 0 / 14, 1 / 37 |
| NRMAX 120 | 13 / 42 | 1020 / 68 |

No run stops and none shows a grid-scale mode: the odd-even part of the beam parallel momentum near the edge is 0.016–0.026 of its maximum at `NRMAX` = 60, the level of the non-uniform mesh, and 0.005 at 120. The steps at `ICMAX` on the circular equilibrium at `NRMAX` = 120 are limited by the impurity diamagnetic flow (LQz8) at the axis node. They do not occur with `FSRP` = 0 and remain when the ripple terms inside $\rho = 0.3$ are removed (tested: $g = 0$ or $D_\mathrm{b}^\mathrm{rp} = 0$ there): it is the known weakness of the impurity equations at the axis, triggered by the state the ripple produces, not a defect of the ripple model.

**Conservation.** Volume integrals of every term of every equation (JT-60U case below, every step): the discrete time derivative equals the sum of the terms to $10^{-6}$ for all density and momentum equations; $n_\mathrm{b} + g n_\mathrm{b}^\mathrm{rp}$ closes with a residual below $10^{-5}$ of the NBI source. The injected torque is 6.73 N m (analytic $2PR_\mathrm{tan}/v_\mathrm{b}$: 6.76 N m). The friction between the beam and the thermal species conserves momentum since the correction of $\ell_{11}^{fk}$ (Sec. 6.5), and the friction terms of all species sum to less than $10^{-8}$ N m at all times since the assembly of Sec. 6.6; before that, the friction between the thermal ions and the impurity left a discrete residual that grew with the rotation to +0.037 N m (0.55% of the injected torque) at 1.5 s. The viscous flux at the wall is zero by the weak form; the rotation is damped by the SOL parallel loss and charge exchange at the edge.

**Ripple benchmark.** JT-60U-like circular case of [1]: $R = 3.4$ m, $a = 0.95$ m, $B = 2.6$ T, $I_\mathrm{p} = 1$ MA, $\kappa = 1.15$ for the ripple, $N = 18$, $\delta_\mathrm{min} = 2 \times 10^{-6}$ (edge ripple 1.26%), 3.2 MW co-tangential and 0.4 MW perpendicular NBI, $\chi = \mu = 0.22(1 + 19\rho^2)$ m$^2$/s, $D = 0.15(1 + 4\rho^2)$ m$^2$/s, 1.5 s:

| case | $u_{\mathrm{i}\phi}(0)$ km/s | $u_{\mathrm{i}\phi}(0.82)$ km/s | $E_r(0.5)$ kV/m | $j_r(0.97)$ A/m$^2$ | $j \times B$ torque N m | $dL_\phi/dt$ N m |
|---|---|---|---|---|---|---|
| no ripple | 658 | 152 | +83 | −0.020 | +0.18 | +0.12 |
| half ripple (FST) | 417 | 60 | +34 | −0.108 | −3.40 | +0.05 |
| full ripple | 128 | −42 | −22 | −0.185 | −7.44 | 0.00 |

An earlier version of this table gave the state at 0.6 s. At that time the runs without ripple and with half the ripple were still transient ($dL_\phi/dt = +1.57$ and +0.87 N m against the injected 6.73 N m; $u_{\mathrm{i}\phi}(0) = 473$ and 325 km/s); the table now gives the quasi-steady state at 1.5 s, where $dL_\phi/dt$ is below 2% of the injected torque in all three runs. With ripple the return current and its $j \times B$ torque changed by 5–11% from 0.6 s to 1.5 s, so the ripple results are qualitatively unchanged; the co-rotation without ripple is 40% higher than at 0.6 s. The return current grows with the ripple and has the magnitude of [1] (−0.08 and −0.16 A/m$^2$ with and without FST),

its torque is counter and grows, $E_r$ moves negative and the edge rotates counter with the full ripple; halving the ripple weakens all of these. The qualitative physics of the 2008 work is kept by B-full and the conservative LQr1. The co-rotation without ripple is 2.2 times that of [1] (300 km/s). The injected torque agrees with its analytic value, so this is a question of the momentum damping and transport (the losses concentrate at the edge in the V5.60 SOL and neutral models), not of the ripple model.

# 7 Models, switches and defaults of V5.60

V5.60 changes the default SOL and neutral physics. The detailed physics, the verification and the development history are in `neutral_transport.tex`; this section states what the code solves.

## 7.1 SOL parallel loss

Outside the separatrix the parallel loss rate is

$$\nu_{\mathrm{L}} = k_L \frac{C_s}{L_c}\, g(\rho), \qquad L_c = 2\pi q R, \qquad g = \frac{x^2}{1+x^2}, \quad x = \frac{r-a}{\delta}\ (r \geq a),\ x = 0\ (r < a), \tag{7.1}$$

and enters every SOL equation in Robin form, e.g. $-\nu_{\mathrm{L}}(n_s - n_{\mathrm{ssup}})$. $k_L$ is `FSLP`. The ramp $g(\rho)$ is numerical smoothing: it keeps $\nu_{\mathrm{L}}$ from jumping at $\rho = 1$, which the FEM cannot represent, and its continuum limit is the step. `MDLSOL` selects its width $\delta$:

| `MDLSOL` | $\delta$ |
|---|---|
| 0 | $3\sqrt{0.5\,T_i(0)\,m_i}\,q(a)/(Z_i e B_\phi(a))/\sqrt{a/R}$ (V5.52) |
| 1 | ion banana width at the separatrix, $\sqrt{\epsilon}\,\sqrt{T_i m_i}/(Z_i e B_\theta)$ (default) |

With `MDLSOL` = 0 the width is evaluated with half the on-axis ion temperature and the edge $q$; for the X-point test equilibrium distributed with the code (`eqdata`, $R = 3.2$ m, $a = 0.8$ m) it is 1.8 times the whole SOL, so $g$ is only 0.17 at the wall and the ramp suppresses the loss everywhere instead of smoothing its onset. `MDLSOL` = 1 evaluates the banana width with the local ion temperature and poloidal field at the separatrix and gives $g \simeq 0.99$ at the wall.

The default $k_L = 1$ is the value of the sheath-limited two-point model (pressure conservation, Mach number 1 at the target, upstream and target densities related by $n_{\mathrm{t}} = n_{\mathrm{u}}/2$) when $2\pi q R$ is read as the length between the two targets: a flux tube of that length loses $2 n_{\mathrm{t}} C_s$ per unit area while holding $n_{\mathrm{u}}\, 2\pi q R$, so $\nu_{\mathrm{L}} = C_s/(2\pi q R)$. V5.52 used the empirical value $k_L = 0.3$, a typical upstream Mach number of the SOL flow; with `MDLSOL` = 1 it does not give a stable SOL on the X-point test equilibrium.

## 7.2 Wall boundary condition for the densities

The table of the previous section lists a Neumann condition for the densities at the wall $\rho = \rho_b$, said to come naturally from the pressure equation. The natural boundary condition of that equation is, however, zero heat flux, not a condition on the density. Up to V5.52 the densities therefore had no condition at the wall, and the radial particle flux through $\rho_b$ was fixed by the last element. On the X-point test equilibrium the ratio of the ion flux through the wall to that through the separatrix, $\Gamma_{\mathrm{wall}}/\Gamma_{\mathrm{sep}}$, was $+0.57$, $-0.40$ and $-0.28$ at `NRMAX` = 60, 120 and 480: depending on the mesh the wall acted as a source or a sink of particles, and refining the mesh did not make it vanish. The radial flux $n_s u_{\mathrm{s}}^V$ cannot take the condition instead: its equation is first order

and not integrated by parts, so it already has its one condition at the axis. For the charged species $\rho_b$ is a computational boundary at the outer edge of the SOL, not a material wall: their exhaust is carried by $\nu_{\mathrm{L}}$.

V5.60 (`MDLWBC` = 1) imposes $dn_s/d\rho = 0$ on $n_{\mathrm{e}}$, $n_{\mathrm{i}}$ and $n_{\mathrm{z}}$. In the P1 basis this is the constraint

$$n_s(N) = n_s(N-1), \qquad N = \texttt{NRMAX}. \tag{7.2}$$

**Constrained degrees of freedom, in general.** For a linear system $Ax = b$ with the constraint $x_N = x_{N-1}$, write $x = Ty$, where $T$ is the identity on $y_1 \dots y_{N-1}$ and copies $y_{N-1}$ into $x_N$. The constrained Galerkin system is

$$T^{\mathrm{T}} A T\, y = T^{\mathrm{T}} b. \tag{7.3}$$

$T^{\mathrm{T}}$ adds row $N$ to row $N-1$; $AT$ adds column $N$ to column $N-1$. In finite elements this is the basis reduction $\phi_{N-1} \to \phi_{N-1} + \phi_N$: the trial and test functions are constant on the last element, which is the P1 form of $dn/d\rho = 0$. The code keeps the system square and solves the equivalent

$$\text{row } N-1 \leftarrow \text{row } N-1 + \text{row } N, \qquad \text{row } N \leftarrow x_N - x_{N-1} = 0, \tag{7.4}$$

with $x_N$ still an unknown, so the other equations keep their coupling to it and the column sum is carried by the constraint row. Since $\sum_i \phi_i = 1$, the sum of all rows of a continuity equation is the global particle balance; each column of $T^{\mathrm{T}}$ sums to one, so that sum, and with it particle conservation, is unchanged. Replacing row $N$ by the constraint without adding it to row $N-1$ would drop the particle balance of the wall half-cell, an $O(h)$ conservation error.

**Implementation.** `txcala.f90`, subroutine `BOUNDARY_ZEROGRAD(LQ)`, called from the boundary-condition block of `TXCALA` for `LQe1`, `LQi1` and `LQz1` after the Dirichlet eliminations. In the band storage the coefficient of term `NC` of equation `NQ` in row `NR` is `ALC`, `BLC` or `CLC(NR,NC,NQ)` for `X(NR+1)`, `X(NR)` or `X(NR-1)`, and `PLC` holds the known terms. For every term `NC = 0..NLCMAX`: `BLC(N-1) += CLC(N)`, `ALC(N-1) += BLC(N)`, `PLC(N-1) += PLC(N)`; row `N` is then cleared and set to `BLC(N,1) = 1`, `CLC(N,1) = -1`. The time term and its right-hand side are built from the same coefficient arrays, so they are folded with them. The condition converges at first order in the mesh size: $\Gamma_{\mathrm{wall}}/\Gamma_{\mathrm{sep}}$ is $-1.8$, $-1.1$, $-0.5$ and $-0.3\%$ at `NRMAX` = 60, 120, 240 and 480.

### 7.3 $C^0$–$C^{5+}$ superstage with a source-slaved $C^0$ part

Carbon is carried by two densities: $n_{\mathrm{z}}$ for $C^{6+}$ and the superstage $\bar{n}_z = \sum_{q=0}^{5} n_{C^{q+}}$, called the bundle below. The bundle exchanges particles with $C^{6+}$ through the effective ionization $f_5 S_5$ and recombination $A_6$ of Sec. 1, and is fed by sputtering in the SOL: $\gamma_{0\mathrm{z}}$ (`rGamm0zz`) is the fraction of the $C^{6+}$ parallel loss and $\gamma_{\mathrm{i,z}}$ (`rGamm0iz`) the fraction of the main-ion parallel loss that return as sputtered carbon.

In local ionization balance the $C^0$ fraction of the bundle is below $2 \times 10^{-6}$, so the bundle is almost entirely magnetized $C^{1+}$–$C^{5+}$ ions. With `MDLNZD` = 1 it is therefore diffused with the anomalous particle diffusivity of the charged species, $D_{nz} = D_{\mathrm{c}}$ (`Dfs`). The exception is the carbon that has just been sputtered, which enters as neutral $C^0$. Its lifetime $1/(S_0 n_{\mathrm{e}})$ is of the order of microseconds, shorter than every other time scale here, so its density follows its local source balance:

$$n_{\mathrm{C0}} = \min\left(\frac{S_{\mathrm{wall}}}{S_0 n_{\mathrm{e}}},\ \bar{n}_z\right), \quad S_{\mathrm{wall}} = \gamma_{0\mathrm{z}}\nu_{\mathrm{L}} \max(n_{\mathrm{z}} - n_{\mathrm{zsup}}, 0) + \gamma_{\mathrm{i,z}}\nu_{\mathrm{L}} \max(n_{\mathrm{i}} - n_{\mathrm{isup}}, 0), \tag{7.5}$$

where $S_0$ is the $C^0 \rightarrow C^{1+}$ ionization rate (OpenADAS scd96, `data/Carbon-S0F0_20260921.txt`). $n_{C0}$ enters the $\bar{n}_z$ equation as a known field through the excess flux with $\Delta D_{C0} = D_{C0} - D_c$, the part of the $C^0$ flux $-D_{C0}\nabla n_{C0}$ that the $D_c$ term does not already carry, so that the bundle flux is exactly $-D_c\nabla(\bar{n}_z - n_{C0}) - D_{C0}\nabla n_{C0}$. Here $D_{C0}$ is the Larsen-limited (see below) value of $\bar{v}_{0z}^2/(3S_0 n_e)$, the short-mean-free-path diffusivity of $C^0$, and $\bar{v}_{0z} = \sqrt{4V_{0z}^2/\pi}$ its mean speed ($V_{0z}$ = `V0z`), and through the parallel loss of the charged part of the bundle, $(1-\gamma_{0z})\nu_L(\bar{n}_z - n_{C0})$, of which the fraction $\gamma_{0z}$ is sputtered back (the $\bar{n}_z$ equations of Secs. 1 and 3). $n_{C0}$ is evaluated from the previous Picard iterate, $S_{wall}$, $S_0 n_e$ and the bound $\bar{n}_z$ alike, so the bound $n_{C0} \leq \bar{n}_z$ applies to the bundle at the time level being solved for, and the $\bar{n}_z$ equation stays linear in $\bar{n}_z$. With `MDLNZD` = 0 (the V5.52 model), $D_{nz} = D_{01}$, the diffusivity of the slow hydrogen neutrals, and the $n_{C0}$ terms are absent.

**Element-based $C^0$ limiter.** $D_{C0}$ is Larsen-limited, $D \rightarrow D/(1 + D|\nabla n_{C0}|/(\alpha \bar{v}_{0z} n_{C0}))$, $\alpha$ = `FLlim`. The limiter is evaluated on each element from the element mean of $n_{C0}$ and the element gradient. It acts on the whole $D_{C0}$, because it bounds the whole $C^0$ flux by free streaming, and $D_c$ is subtracted afterwards with its sign: $\Delta D_e = D_e - D_{c,e}$, where $D_e$ is the limited $D_{C0}$ and $D_{c,e}$ the element mean of $D_c$. $\Delta D_e < 0$ is not a negative diffusion: it removes the part $D_{c,e}$ of the $C^0$ flux that the $D_c$ term already carries, and since $n_{C0}$ is a known field the matrix is unchanged. It occurs in the first element outside the separatrix, where $n_{C0}$ rises from zero within one element and the limited $D_{C0}$ falls to about $\alpha \bar{v}_{0z} h/2$ ($h$ the radial width of the element): at NRMAX = 480, not at NRMAX $\leq$ 240. Clipping $\Delta D_e$ at zero there would let the $C^0$ flux exceed free streaming by 7%. The element flux $F_e = \langle|\nabla V|^2\rangle_e \Delta D_e \, \Delta n_{C0}/\Delta V_e$ enters the two rows of the element as $+F_e$ and $-F_e$, so the term is exactly conservative. The element evaluation is needed because inside the separatrix the source, and hence $n_{C0}$, vanishes: a nodal ratio $|\nabla n_{C0}|/n_{C0}$ would be 0/0 at the separatrix node, and whether that node is limited would be left to rounding. The limiter is part of the model and is always applied with `MDLNZD` = 1. Without it $D_{C0}$ becomes very large in the SOL, and the divergence of the known $C^0$ flux, a source that does not scale with $\bar{n}_z$, drives $\bar{n}_z$ negative just outside the separatrix.

**Validity of the source-slaved $C^0$ part.** The slaving of $C^0$ to its source is implemented and is part of the production model. What remains open is the range of validity of this quasi-steady approximation and the lack of a formal positivity guarantee. The mean free path of $C^0$ is comparable to the width of the $C^0$ layer, so the local source balance does not resolve the inner structure of the layer. Positivity of $\bar{n}_z$ is kept by the limiter in the cases computed, but it is not guaranteed, because the divergence of the known $C^0$ flux is a source that does not scale with $\bar{n}_z$. The bound $n_{C0} = \bar{n}_z$ marks the limit of the model. It is never reached in the converged iterates of the production cases; it is reached only in the first sweep of a run started from the initial profiles, where $\bar{n}_z$ is still close to zero. Where it holds persistently, the lagged treatment of $n_{C0}$ is not appropriate: the $C^0$ flux in the bounded nodes is then a diffusion of $\bar{n}_z$ itself with $\Delta D_{C0} \approx 7$–$15\, D_c$, taken from the previous iterate, and the iteration diverges (forced with a $C^0$ ionization rate reduced 100-fold; with the bound taken from the previous time step, the time stepping diverges instead). `TXLOOP` warns when the bound is still active in the last Picard sweep of a step, and with `IDIAG` $\geq 2$ prints the number of bounded nodes at every step. Such conditions call for an independent $C^0$ transport equation, an implicit treatment of the $C^0$ flux, or a kinetic (Monte-Carlo) treatment, which would also delimit the validity of the approximation.

### 7.4 Larsen limiter on the hydrogen slow neutrals (optional)

The slow hydrogen neutrals are diffused with the short-mean-free-path diffusivity $D_{01} = \bar{v}_0^2/(3S_{\rm tot}) = \lambda\bar{v}_0/3$, $S_{\rm tot}$ being the ionization plus charge-exchange rate and $\lambda = \bar{v}_0/S_{\rm tot}$ the mean free path. The Larsen limiter replaces $D$ by $D/(1+R)$ with

$$R = \frac{D|\nabla n_{01}|}{\alpha\,\bar{v}_0\,n_{01}} = \frac{\text{flux the diffusion law demands}}{\text{free-streaming flux it may not exceed}} = \frac{\lambda}{3\alpha L_n}, \qquad \alpha = \texttt{FLlim},$$

$L_n$ being the gradient length: the limited flux $D|\nabla n|/(1+R)$ is the diffusion flux where $R \ll 1$ and $\alpha\bar{v}_0 n$ where $R \gg 1$. $\alpha = 1/4$ is the one-sided flux of an isotropic population: for particles of speed $\bar{v}$ spread uniformly over all directions, those crossing a plane in one direction carry $n\bar{v}\int_0^1 \mu\,d\mu/2 = n\bar{v}/4$. $\alpha = 1/2$ would correspond to all particles moving into one half-space, uniformly in angle, and $\alpha = 1$ to all of them moving in the same direction, the kinematic upper bound. In the V5.60 solution without the limiter $R$ is 1–3 in the SOL, about 2 at the separatrix and up to 6 at $\rho \simeq 0.8$–$0.9$ (Kn $= \lambda/L_n \sim 1$): the diffusion approximation is marginal wherever the neutrals matter.

`MDLNOFL` $= 0$ (default) leaves $D_{01}$ unlimited. `MDLNOFL` $= 2$ evaluates the limiter element by element, as for the $C^0$ flux, from the element mean $\bar{n}$ and the element difference $\Delta n$ of the current iterate; since $|\Delta n|/\bar{n} \le 2$ on an element, the factor is bounded by $1 + 2\lambda/(3\alpha h)$. `MDLNOFL` $= 1$ is the earlier nodal form, with $|\nabla n|/n$ from the spline derivative, kept only to reproduce earlier runs. The limited ionization front at $\rho \simeq 0.85$–$0.9$ is exponential with the decay length $\alpha\lambda \simeq 5$ mm, shorter than an element at `NRMAX` $= 60$ (about 1 cm). On that unresolved front, and in the tiny-$n_{01}$ tail inside it, the nodal ratio is large and erratic, and the Picard iteration does not settle: with `MDLNOFL` $= 1$, 496 of 500 steps reach `ICMAX` at `NRMAX` $= 60$ (`EPS` $= 10^{-6}$), the neutral-only relaxation `RLXNO` $= 0.5$ (used only with `MDLNOFL` $= 1$) leaves 2–3. `MDLNOFL` $= 2$ needs no relaxation: 20.0 / 21.7 sweeps on average and no step at `ICMAX` (X-point / circular, `NRMAX` $= 60$), the same as without the limiter. The two forms agree to 0.5–2% where both converge. Against the limiter off, the limiter raises the neutral inventory by about 30% and changes $n_{\rm e}$ at the separatrix and at the wall by $-3\%$ and $+4\%$; $Z_{\rm eff}(0)$ is unchanged.

The limiter is a bound on the diffusion flux, not a substitute for kinetic neutral transport. A straight-flight reference (the same sources, attenuated at $S_{\rm tot}$) ionizes or charge-exchanges 35% of the slow neutrals inside the separatrix (X-point, `NRMAX` $= 240$); plain diffusion gives 49% and the limiter with $\alpha = 1/4$ 25%, with almost nothing beyond $\rho = 0.95$ against 12% in the reference. $\alpha = 1/2$ recovers the total but not the ballistic tail, and no single $\alpha$ recovers both. The difference between limiter on and off is of the size of the uncertainty from the neutral energy alone (the reference gives 24% and 48% at 0.5 and 4.5 eV). The limiter is therefore off by default.

### 7.5 Anderson acceleration of the Picard iteration (optional)

Each time step is solved by Picard iteration: the coefficients are evaluated from the previous iterate $x_k$ and the linear system gives $g_k = G(x_k)$. On fine meshes the iteration has a slow mode localised on the first few nodes at the magnetic axis, in $BV_{1s}$ (`LQ*8`) of all species, whose contraction factor per sweep tends to 1 as the axis element shrinks (0.70, 0.91, 0.98 at `NRMAX` $= 120$, 240, 480 with $\Delta t = 10^{-4}$); at `NRMAX` $= 480$ it takes about 500 sweeps to reach `EPS` $= 10^{-6}$.

With `NANDER` $\neq 0$ the next iterate is instead

$$x_{k+1} = g_k - \sum_j \gamma_j\,(\Delta x_j + \Delta f_j), \qquad \gamma = \arg\min \Big| f_k - \sum_j \gamma_j \Delta f_j \Big|_2,$$

with $f = g - x$ and $\Delta x_j$, $\Delta f_j$ the differences over the last |NANDER| sweeps of the step (Anderson mixing with $\beta = 1$). Each variable is scaled by its profile RMS at the first sweep of the step, as in the convergence test. With NANDER $< 0$ the rows of LQm1–LQm5 do not enter the least-squares fit, because near convergence their increments are rounding (for $\dot{\psi}_t$ the floor of Sec. 7.6); the correction is applied to all variables. The history is restarted at every time step, and dropped (the plain $g_k$ is used) if the least-squares solve fails or returns $|\gamma| > 10^3$. Once the adaptive relaxation of the step has found the iteration not contracting, the rest of the step is left to the relaxation alone: the linear model behind the mixing does not hold there, and with the mixing kept on the first steps from the initial profiles at NRMAX = 960 end in a negative density. A fixed point of $G$ is a fixed point of the accelerated iteration, so the converged solution is that of plain Picard; NANDER = 0 (default) is bit-identical to the plain iteration.

With NANDER $= -20$ (100 steps from $t = 0.02$, $\Delta t = 10^{-4}$, EPS $= 10^{-6}$, ICMAX = 100, X-point equilibrium) the average number of sweeps falls from 22.7, 36.4, 99.4 to 16.0, 28.4, 58.8 at NRMAX = 60, 120, 240, and the steps that reach ICMAX at NRMAX = 240 from 86 to 9; the circular equilibrium behaves alike. The solution agrees with the plain iteration to within the convergence tolerance. At NRMAX = 480 the LQ*8 mode is removed as well. These figures were measured with the purely relative test of $\dot{\psi}_t$ (LQm2), under which almost every step at NRMAX = 480 still reached ICMAX on the rounding floor of $\dot{\psi}_t$, which the acceleration does not affect. With the floor term of Sec. 7.6 the steps at NRMAX = 240 and 480 end when the other variables have converged: with NANDER $= -20$ the average number of sweeps is 33.5 (240) and 67.1 (480), and 3 of 100 steps at 480 reach ICMAX, on the other variables.

### 7.6 Convergence test of $\dot{\psi}_t$ (LQm2)

The sweeps of a time step stop when, for every variable $X_q$, $\max_j |\Delta X_{q,j}| \le \mathtt{EPS}\,\mathrm{rms}(X_q)$, $\Delta X_q$ being the change over the sweep. For $X_2 = \dot{\psi}_t$ this relative test has a floor that no iteration removes, and the test is

$$\max_j |\Delta X_{2,j}| \le \mathtt{EPS}\,\mathrm{rms}(X_2) + C_\mathrm{r}\,\epsilon_\mathrm{m}\,\mathtt{NRMAX}^2\,\frac{\mu_0\,\mathrm{rms}(X_5)}{\mathtt{ADV}\,\Delta t}, \qquad C_\mathrm{r} = 1,$$

with $X_5 = \psi_t/\mu_0$ (LQm5), $\epsilon_\mathrm{m}$ the machine epsilon and ADV the weight of the implicit terms (2/3 for BDF2, 1 in the first step). LQm2 is not removed from the test: the relative term is kept as it is for every variable, and only the floor term is added. The test of every other variable (including $\dot{\psi}$, LQm3), the equations and the fixed point are unchanged.

The floor has two factors. $\psi_t$ is mostly vacuum flux, and in Ampère's law (the LQm2 row) the $\nabla \times B_\phi$ term applied to it is a cancellation of terms larger than the result, $|KX_5|/(|K||X_5|) = 3.4 \times 10^{-7}$ at NRMAX = 60 falling as NRMAX$^{-2}$. A rounding of relative size $\epsilon_\mathrm{m}$ in that row, in the assembly or in the solve, leaves $\psi_t$ determined only to $(0.05\text{–}0.07)\,\epsilon_\mathrm{m}$NRMAX$^2$. Faraday's law (the LQm5 row) then gives $\dot{\psi}_t$ as the difference quotient $\mu_0(X_5 - X_5^\mathrm{old})/(\mathtt{ADV}\,\Delta t)$, which multiplies that rounding by $\psi_t/(\mathtt{ADV}\,\Delta t\,|\dot{\psi}_t|)$, $7 \times 10^6$ at $\Delta t = 10^{-4}$, $t = 0.02\,\mathrm{s}$, and without bound as $\psi_t$ approaches a steady state. The relative floor of $\dot{\psi}_t$ is therefore about $4 \times 10^{-7}$, $1.5 \times 10^{-6}$, $4.6 \times 10^{-6}$, $1.8 \times 10^{-5}$ at NRMAX = 60, 120, 240, 480 ($\Delta t = 10^{-4}$, $t = 0.02\,\mathrm{s}$), grows as $1/\Delta t$, and grows in time: at NRMAX = 60 it reaches $10^{-5}$ by $t = 0.5\,\mathrm{s}$. At the floor the equations are satisfied to rounding (backward error of the LQm2 and LQm5 rows $10^{-15}$–$10^{-11}$, normwise residual of every equation below $4 \times 10^{-9}$); the increment of $\dot{\psi}_t$ is the forward error of an ill-conditioned component, not an unconverged iteration. The second term of the test is that floor with a safety factor.

$C_\mathrm{r}$ is not a physical tolerance on $\dot{\psi}_t$. $C_\mathrm{r} = 0.3$ lies on the largest floor observed (the rounding of $\psi_t$ reaches 4–6.5 times its geometric mean) and leaves no margin for rounding that depends on the compiler, the BLAS

or the order of operations; $C_\mathrm{r} = 1$, three to four times it, is used. The two values give the same numbers of sweeps to within a few per cent and solutions that cannot be told apart. The test on $\dot{\psi}_t$ keeps its role while the iteration converges, since $\dot{\psi}_t$ responds to a change of the currents about $10^4$ times more strongly than they change (`NRMAX` = 60). In the printed and returned convergence measures the value for `LQm2` is the part of the relative increment above the floor term.

Against a reference solved with the relative test and up to 300 (100 steps) or 200 (3000 steps) sweeps per step, `NANDER` = −20, the test with the floor term gives (`EPS` = $10^{-6}$, `ICMAX` = 100, $\Delta t = 10^{-4}$)

| | relative test<br>sweeps / steps at `ICMAX` | with the floor term<br>sweeps / steps at `ICMAX` |
|---|---|---|
| X-point, 240, 100 steps | 58.8 / 9 | 33.5 / 0 |
| circular, 240, 100 steps | 69.8 / 10 | 37.1 / 0 |
| X-point, 480, 100 steps | 99.6 / 97 | 67.1 / 3 |
| circular, 480, 100 steps | 99.3 / 95 | 77.0 / 5 |
| X-point, 240, 3000 steps | 89.0 / 2167 | 29.9 / 0 |
| circular, 240, 3000 steps | 89.6 / 2093 | 33.6 / 0 |

The remaining steps at `ICMAX` are steps on which another variable has not converged. Densities, temperatures, current and $Z_\mathrm{eff}$ differ from the reference by less than $7 \times 10^{-8}$; $\dot{\psi}_t$ and the poloidal electric field derived from it by 0.2–$1.5 \times 10^{-4}$, within the 0.3–$2.1 \times 10^{-4}$ by which the relative test with `ICMAX` = 100 differs from the same reference, that is, within the floor. The loop voltages and the Poynting-flux terms follow within $6 \times 10^{-7}$ over 3000 steps, except the rate of change of the poloidal electric-field energy, which is built on $\epsilon_0 \dot{\psi}_t{}^2$ and carries the floor of $\dot{\psi}_t$ under either test. Volume-averaged densities, stored energy and current agree to $10^{-10}$ and the toroidal angular momentum to $2 \times 10^{-8}$. The BDF2 errors and orders of $n$, $T$, $j$ and $Z_\mathrm{eff}$ are unchanged (`NRMAX` = 60, $\Delta t = 4 \times 10^{-4}$–$2.5 \times 10^{-5}$, `EPS` = $10^{-8}$); only the self-convergence error of $\dot{\psi}_t$ between the two finest steps rises from 6–$8 \times 10^{-6}$ to $1.5 \times 10^{-5}$, its floor. With `NANDER` = 0 the slow `LQ*8` mode still decides most steps at `NRMAX` ≥ 240, and the floor term changes little.

### 7.7 New defaults and V5.52 compatibility

| parameter | V5.52 | V5.60 | meaning |
|---|---|---|---|
| `FSLP` | 0.3 | 1.0 | $k_L$ |
| `MDLSOL` | (0) | 1 | ramp width $\delta$ |
| `MDLNZD` | (0) | 1 | bundle transport with the source-slaved $C^0$ part |
| `MDLWBC` | (0) | 1 | wall density condition |
| `MDLNOFL` | (0) | 0 | Larsen limiter on $D_{01}$ (2: element form; 1: legacy nodal form) |
| `FLlim` | – | 0.25 | limiter factor $\alpha$ ($C^0$; $D_{01}$ with `MDLNOFL` = 1, 2) |
| `RLXNO` | – | 0.5 | neutral-only relaxation (legacy `MDLNOFL` = 1 only) |
| `NANDER` | (0) | 0 | Anderson acceleration of the Picard sweeps |

A value in parentheses means that the switch did not exist in V5.52 and that the value reproduces the V5.52 behaviour. These are all the defaults that differ. `parm/txparm.v552default` sets `FSLP` = 0.3, `MDLSOL` = `MDLNZD` = `MDLWBC` = `MDLNOFL` = 0. With it V5.60 is not bit-identical to V5.52.20 (the optimized build schedules the changed routines differently), but one step from the same state gives double-precision states that agree to $10^{-12}$ (median over the variables) and $10^{-9}$ at worst, except $\dot{\psi}_t$ at its Picard noise floor ($3 \times 10^{-7}$); the profile

output agrees to its single-precision resolution. The table `data/Carbon-S0F0_20260921.txt` is read at start-up in either case.

### 7.8 Other changes since V5.51

- `MDLPCK` = 2 (default): block-tridiagonal solver with one step of iterative refinement; the Picard floor falls from about $10^{-5}$ to $5 \times 10^{-7}$ on the worst equation.
- `oldmixa` (default 0.5): Picard under-relaxation applied only on time steps whose sweep is measured not to contract.
- BDF2 keeps its history across `CONT`; one backward-Euler step is taken where there is none (cold start, `LOAD`).
- Analytic axis limit of the Pfirsch–Schlüter Fourier factors, $f_m(\cdot, 0) = 0$; $\hat{q}^2$ at the axis from the interior instead of $q(0)^2$.
- $D_{0\mathrm{z}}$ uses the impurity neutral speed $V_{0z}$.
- `FSTPTM` = 3 (default): multi-ion composition channel for the turbulent particle flux. With `FSTPTM` = 2 the impurity is not diffused on its own density gradient, and in long runs a grid-scale mode of $n_\mathrm{z}$ (wavelength about three mesh intervals, sign-alternating) grows: the JT-60U case of Sec. 6.7 stops at 2.4 s with $T_\mathrm{z} < 0$ at $\rho \approx 0.43$. The mode is tied to the mesh, not to a physical scale, and is not a physical impurity instability; it is the discrete form of an under-regularized closure, in which nothing damps short-wavelength perturbations of $n_\mathrm{z}$. The diffusion of each thermal ion on its own gradient in `FSTPTM` = 3 damps them at the rate $Dk^2$; the same case then runs to 4 s with no grid-scale component above the curvature of the profile. In the NBI cases $n_\mathrm{e}$ and $u_{\mathrm{i}\phi}$ change by less than 1.5% (in the ohmic default case $u_{\mathrm{i}\phi} \approx 0.5$ km/s changes by 15%), the core impurity density by 25–55%. `FSTPTM` = 2 remains available and must be given for every species (`FSTPTM` = 2, 2, 2).
- Turbulent force of `FSTPTM` = 2, 3 in the wall row. The pressure-gradient part of the electron-driven turbulent force is assembled through the diamagnetic flow ($n\,BV_1$, LQ*8) and $n\Phi'$. LQe8 is the algebraic definition $n_\mathrm{e} BV_{1\mathrm{e}} = -(I/\psi')(p_\mathrm{e}'/(Z_\mathrm{e}e) + n_\mathrm{e}\Phi')$, so in the continuum this is the direct $\partial_V p_\mathrm{e}$ and the force is `FSVAHL` $\partial_V p_\mathrm{e}$ + (1 − `FSVAHL`) $T_\mathrm{e}\partial_V n_\mathrm{e}$. Discretely LQ*8 is solved with the SUPG weights $N_j + phN_j'$ (`iSUPG8` = 1) and the force rows with the Galerkin $N_i$: at an interior node the SUPG moments of the two elements around it cancel to $O(h^2)$, but the wall node has the last element only, and there the two forms differ by the order of the LQe8 residual. The particle flux at the wall is fixed by the wall row of LQ*4, so this gave a spurious wall flux (JT-60U case, `NRMAX` = 60: an electron inflow of $0.22 \times 10^{20}$ s$^{-1}$). The wall row (test function of node `NRMAX`) now uses the direct-gradient form for every species (that of `FSTPTM` = 0); not with `MDFIXT` ≠ 0. The removed artificial part of the electron wall flux is 0.38 and 0.20 $\times 10^{20}$ s$^{-1}$ at `NRMAX` = 60 and 120 (1.3 s) and 0.16, 0.10 and 0.045 at 60, 120 and 240 (0.05 s), so it goes with the mesh; the corrected wall flux itself converges slowly (+0.16 and +0.05 at 60 and 120; +0.18, +0.03 and +0.02 at 60, 120 and 240), being a small difference of large terms in the wall row. The non-ambipolar part of the wall flux (the fast-ion return current), $j_r$, the $j \times B$ torque and $L_\phi$ are unchanged to three digits, the profiles change by less than 0.5% (JT-60U case to 4 s, `FSRP` = 0 and 1, `FSTPTM` = 2 and 3, X-point and circular test equilibria), and the Picard counts are unchanged.

- `FSTPTM` = 4 (option; the default stays 3): thermal-species self-gradient closure. With fast ions, `FSTPTM` = 3 drives the common transport by $\nabla n_\mathrm{e}$, which contains the fast-ion charge gradient, and lets the main ions carry the fast-ion share of the electron flux (weight $n_\mathrm{i} + n_\mathrm{b}$), so the main-ion continuity equation has a sink $\nabla \cdot (n_\mathrm{b} u)$ that does not vanish with $n_\mathrm{i}$. `FSTPTM` = 4 drives the electrons by the thermal-ion charge density $n_\mathrm{th} = \sum_s Z_s n_s$, gives the reaction to the thermal ions with the weights $Z_s n_s / n_\mathrm{th}$ and uses $n_\mathrm{th}$ in the composition pairs, so that every thermal species diffuses on its own density gradient with the common thermodiffusion, $\langle \mathbf{\Gamma}_s \cdot \nabla V \rangle = -D n_s (X_s + C_\mathrm{T} X_T)$, also with fast ions, and the fast ions have no turbulent flux. Momentum and ambipolarity hold element by element (the weights sum to one by construction). Without fast ions it equals 3 (JT-60U ohmic, 300 steps: $10^{-7}$ in the densities, $5 \times 10^{-5}$ in $u_{\mathrm{i}\phi}$, the quasineutrality residual); it matches the diagnostic implementation it was developed in to $5 \times 10^{-8}$; `FSTPTM` = 3 results are bit-identical. The formulation is in twofluid.tex. It must be given for every species and needs `MDFIXT` = 0; `NCM` and the save file are unchanged. Known limits: the beam friction (NBCD) and the beam radial flux still reach the thermal ions through ambipolarity, so positivity of $n_\mathrm{i}$ is not guaranteed structurally; the more peaked core densities expose the axis `LQ*8` Picard stall below already at `NRMAX` = 60. JT-60U case at 1.5 s (`NRMAX` = 60), 4 against 3: $n_\mathrm{e}(0)$ +17% (`PNsSUP(1)` = 0.0125) and +21% (0.01), $n_\mathrm{i}(0)$ +26% and +36%, $n_\mathrm{b}/n_\mathrm{i}(0)$ 0.31 → 0.21 and 0.42 → 0.26, $u_{\mathrm{i}\phi}(0)$ −11% and −14%; with 6.4 MW NBI $n_\mathrm{i}(0)$ doubles ($n_\mathrm{b}/n_\mathrm{i}(0)$ 0.92 → 0.35); $n_\mathrm{e}$ at $\rho = 0.95$ within 3% in all cases, and $\sum_s Z_s \Gamma_s$ inside the separatrix below $1.5 \times 10^{-7}$ of the electron flux. In the high-density cases (6.4 MW NBI, `Zeffin` = 2.5, gas puff 0.4) 216–722 steps of 15000 stall in short bursts (208 in the benchmark with 0.0125), none with `NRETRY` = 1, which changes the profiles by less than $10^{-6}$.

- `NRETRY` = 1 (option, default 0): a step whose Picard iteration reaches `ICMAX` without converging is taken again as two steps of $\Delta t/2$ (backward Euler, then BDF2; the history is $X(t - \Delta t)$ again afterwards). Converged steps are unchanged, and 0 is bit-identical to before. It is meant for sporadic stalls: from a stalled state of the JT-60U case at `NRMAX` = 120 it leaves no unconverged step in 40 (4 without it), but at 240 nearly every step is retried and some still do not converge, so there the time step itself has to be smaller.

- Known numerical limit, the axis `LQ*8`: the Picard map has a nearly neutral mode on the first axis nodes of $BV_1$ (mostly the impurity), fed by the lagged density and flow coefficients together (no single term dominates; the lagged projection of the toroidal viscous momentum flux onto the parallel flow equations, `FSPARV`, stabilizes it). In the JT-60U case its eigenvalue is 1.002 at `NRMAX` = 120 and 1.043 at 240, and a time step proportional to the mesh size restores convergence ($\Delta t \leq 10^{-4} \times 60/$`NRMAX` s, roughly); at `NRMAX` = 240 an odd-even mode of $BV_{1\mathrm{z}}$ and $n_\mathrm{z}$ grows on the axis within 0.1 s at $\Delta t = 10^{-4}$ s. The continuum solution is regular. Use the smaller time step at high `NRMAX`, and `NRETRY` = 1 for the occasional stalls with strongly peaked core densities.

- Recycling at the main wall. The radial ion outflow through the wall $\rho_b$, $\Gamma_s(\rho_b)^+ = \max(n_s \langle \boldsymbol{u}_s \cdot \nabla V \rangle, 0)$ at `NRMAX`, returns as a wall-side inflow of neutrals like the gas puff: `RWALL` (default 0.98) of the bulk ions to $n_{01}$, `RWALLZ` (0.42) of the impurity ions to the impurity neutrals, and `RWALLIZ` (0, off) of the bulk ions to the impurity neutrals by sputtering; the impurity never returns as a hydrogen neutral. This path is separate from the target recycling of the SOL parallel loss (`rGamm0`, `rGamm0zz`, `rGamm0iz`), which `tx_sol_particle_balance` computes from a divertor recycling loop (escape to the upstream SOL against the pump, $c_1/(c_1 + c_5) = 0.980$ for the bulk ions; for the impurity the reflection and self-sputtering yield

0.42 times an escape factor 0.051, 0.0213; sputtering by the bulk ions 0.0099). There is no such loop at the main wall, so `RWALL` is that of a saturated wall with a small retention and `RWALLZ` the yield itself. In the JT-60U case the main wall returns $0.40\times10^{20}$ s$^{-1}$ (target recycling 14.4, gas puff 28.1); $n_{\rm e}$ changes by 0.2% (0.6% with `FSRP` = 1), and even `RWALLIZ` = 0.15 (tested with `RWALLZ` = 0.02) changes the core impurity density by 0.04%, the wall source being screened by the SOL. `W` (TXSTAT) and `O 7` (`tx_sources.dat`) print the SOL parallel loss, the wall fluxes and the neutral sources by path (target, main wall, gas puff) with the ionization. The three coefficients are kept in the save file in a second extension record; older files load with the current values and newer files load in older builds.

- Finite-collisionality closure for the higher-order flows; triangularity of grid-traced equilibria regularized; G-EQDSK $p'$ normalization corrected; `ipbtdir` = 1 by default.

- `MDLNSUP` (default 1): the SOL upstream densities of the ions and the impurity, `PNsSUP(2:3)`, are derived from `PNsSUP(1)` and `Zeffin` (quasi-neutral, $Z_{\rm eff}$ = `Zeffin`) whenever the parameters are read; before, they were set once from the defaults and a namelist value of `PNsSUP(1)` or `Zeffin` left them unchanged. `MDLNSUP` = 0 uses `PNsSUP(1:3)` as given. The SOL loss $-\nu_L(n_s - n_{{\rm SUP},s})$ holds the SOL density near $n_{{\rm SUP},s}$, so `PNsSUP(1)` acts as the upstream SOL density and sets the density level of the whole plasma.

- Diagnostic switches, off by default: `iaxproj` (near-axis regularity projection), `Dnfac` (particle diffusion in the impurity continuity equation).

## References


[1] M. Honda, T. Takizuka, A. Fukuyama, M. Yoshida and T. Ozeki, Nucl. Fusion **48** (2008) 085003.

[2] M. Yoshida et al., Plasma Phys. Control. Fusion **48** (2006) 1673.

[3] M. Kikuchi and M. Azumi, Plasma Phys. Control. Fusion **37** (1995) 1215, Appendix.

[4] T. Tala et al., Proc. 24th IAEA Fusion Energy Conf. (San Diego, 2012) ITR/P1-19.

[5] T. Tala et al., Nucl. Fusion **51** (2011) 123002.

[6] P. Mantica et al., Phys. Plasmas **17** (2010) 092505.

[7] H. Weisen et al., Nucl. Fusion **52** (2012) 042001; **52** (2012) 114024.

[8] M. Honda et al., Nucl. Fusion **53** (2013) 073050.

[9] I. Casiraghi et al., Nucl. Fusion **61** (2021) 116068.